Chapter 19 — Cheat Sheet and Glossary

This chapter is the book on one table. Chapters 1–18 teach each topic with runnable projects; this chapter **looks things up**. Any word you meet here and don't know — starter, BOM, proxy, propagation, slice — has a plain-English entry in the A–Z glossary at the end (§19.16). You come here when you remember that something exists but not its exact name, default or package: "which starter brings `@WebMvcTest` in Boot 4?", "what does `REQUIRES_NEW` do when the caller rolls back?", "which property turns on virtual threads?", "is it `hasRole("ADMIN")` or `hasRole("ROLE_ADMIN")`?".

Every row links to the section that explains it, so a sheet is never the end of the road: if a line is not enough, follow the **§** link. Nothing here is new material; where a sheet quotes a default or a behaviour, a test in `code/cheatsheet-checks/` proves it (§19.15).

What you'll find

1. Versions of the Boot 4.1.1 stack at a glance (§19.2).
2. **Annotations**: every common one → module / starter → purpose → section (§19.3).
3. **Starters** in Boot 4.1, including renamed, deprecated and removed ones (§19.4).
4. **Properties** with their defaults: web, data, cache, security, actuator, observability, Kafka, threads, HTTP clients, API versioning (§19.5).
5. `@Transactional`: attributes, **propagation**, isolation and **rollback** tables (§19.6).
6. **Bean scopes** and the **lifecycle order** (§19.7).
7. **Spring Security 7 DSL** snippets that compile: form login, JWT resource server, method security, CORS/CSRF (§19.8).
8. **Testing** annotations and slices (§19.9), **Reactor** operators (§19.10), **Actuator** endpoints (§19.11).
9. **Which tool for which job** decision tables (§19.12) and a **Boot 3 → 4 checklist** (§19.13).
10. An **interview index** into Chapter 18 (§19.14), the **proofs** (§19.15) and an **A–Z glossary** of more than 500 terms (§19.16).

Versions: Spring Boot **4.1.1** → Spring Framework **7.0.9**, Spring Security **7.1.1**, Java **25** (17 minimum). "(Boot 4)", "(F7)" and "(S7)" mark things that are new or changed in Boot 4, Framework 7 or Security 7.

19.1 How to use this chapter

How the sheets are organised

You want to...	Go to
find the name, jar or purpose of an annotation	§19.3
add a dependency	§19.4
set or look up a property and its default	§19.5
pick a propagation / isolation / rollback rule	§19.6
remember the bean lifecycle order	§19.7
copy a security configuration	§19.8
choose a test annotation	§19.9
find a Reactor operator	§19.10
know what an Actuator endpoint shows	§19.11
choose between two technologies	§19.12
upgrade a Boot 3 app	§19.13
revise for an interview	§19.14
look up a word	§19.16

Conventions

- **Module / starter**: the jar (Java archive — a zip file of compiled classes) that contains the class, and the Boot starter that brings it. "core" means `spring-context` / `spring-beans`, which every starter includes.
- **§N.M** links go to the teaching section of Chapters 1–18. Follow them for the explanation, the trade-offs and the proof.
- Defaults quoted in these sheets come from the Boot 4.1.1 configuration metadata (the `spring-configuration-metadata.json` files inside the Boot jars) or from the chapter that measured them. The ones marked ✔ **proved** are asserted by a test in §19.15.
- Properties are written in `.properties` form. The same keys work in YAML (an indentation-based configuration format) and as environment variables (named values set outside the app by the shell, OS or container) (§3.4).

19.2 Versions at a glance

What `spring-boot-starter-parent` **`4.1.1`** puts on your classpath — the list of jars the JVM loads classes from — (from the `spring-boot-dependencies-4.1.1` BOM — bill of materials, a list that pins compatible versions of many libraries). You choose one number, the Boot version, and Boot chooses the rest (§1.4).

Component	Version	Notes
Java	17 minimum, 25 recommended	this book: Java 25
Spring Framework	7.0.9	Jakarta EE 11 (Servlet 6.1, JPA 3.2, Bean Validation 3.1)
Spring Security	7.1.1	Authorization Server now inside Spring Security (§11.14)
Spring Data	2026.0.1 release train	JPA, JDBC, MongoDB, Redis, R2DBC, REST
Hibernate ORM / Validator	7.4.5 / 9.1.3	
Jackson	3.1.5 (`tools.jackson.*`)	annotations still `com.fasterxml.jackson.annotation` (§5.5)
Tomcat / Jetty	11.0.24 / 12.1.12	Undertow removed
HikariCP / H2 / PostgreSQL JDBC	7.0.2 / 2.4.240 / 42.7.13	
Reactor	2025.0.7 train (Reactor Core 3.8)	
Micrometer / Micrometer Tracing / OpenTelemetry	1.17.1 / 1.7.1 / 1.62.0	
Spring for Apache Kafka / kafka-clients	4.1.1 / 4.2.1	
Spring AMQP	4.1.1	
Flyway / Liquibase	12.4.0 / 5.0.3	Liquibase 5 Community is FSL-licensed (§9.4)
JUnit / Mockito / AssertJ / Awaitility	6.0.3 / 5.23.0 / 3.27.7 / 4.3.0	
Testcontainers	2.0.5	new artifact names `testcontainers-<module>` (§12.14)
Caffeine / Lettuce	3.2.4 / 7.5.2	
Not managed by Boot		import their own BOM
Spring Cloud	2025.1.x "Oakwood" (2025.1.2+ for Boot 4.1)	§16.2
Spring AI	2.0.x (2.0.1)	§17.1
springdoc-openapi	3.x (3.1.1)	§5.12
Resilience4j	`resilience4j-spring-boot4` 2.4.0	§16.8

19.3 Annotations cheat sheet

The annotations of this book, grouped by area. "Needs" names the switch without which the annotation silently does nothing — the most common reason an annotation "doesn't work".

Core container and beans

Annotation	Module / starter	Purpose	§
`@Component`	core	generic bean found by component scanning	§2.3
`@Service`	core	`@Component` for business logic (no extra behaviour)	§2.3
`@Repository`	core	`@Component` for data access + exception translation to `DataAccessException`	§2.3, §7.5
`@Controller` / `@RestController`	spring-web / `-webmvc`, `-webflux`	web handler; `@RestController` = `@Controller` + `@ResponseBody`	§5.3
`@Configuration`	core	class of `@Bean` methods; full mode = CGLIB subclass (`proxyBeanMethods = true`)	§2.3
`@Bean`	core	factory method registering its return value as a bean; name = method name	§2.3
`@ComponentScan`	core	scan a package tree for components (inside `@SpringBootApplication`)	§1.8
`@Import`	core	pull in another configuration class, selector, registrar or `BeanRegistrar`	§2.3
`@Autowired`	core	inject here; optional on a single constructor	§2.4
`@Qualifier`	core	pick a bean by name/qualifier at the injection point (beats `@Primary`) ✔	§2.5
`@Primary`	core	default bean when several match ✔	§2.5
`@Fallback`	core (Framework 6.2)	use this bean only if no other candidate qualifies	§2.5
`@Order`	core	order in `List<T>` injection, aspects, filters, listeners (lower first) ✔	§2.5
`@Value`	core	inject one value: `${key:default}`, SpEL `#{...}` ✔	§3.5
`@Scope`	core	`singleton` (default), `prototype`, web scopes; `proxyMode` for scoped proxies	§2.6
`@RequestScope` / `@SessionScope` / `@ApplicationScope`	spring-web	web scopes with a CGLIB scoped proxy	§2.6
`@Lookup`	spring-beans	method injection: returns a fresh prototype on each call	§2.6
`@Lazy`	core	create on first use; on an injection point, inject a lazy proxy (breaks cycles)	§2.10, §2.9
`@PostConstruct` / `@PreDestroy`	`jakarta.annotation`	init / destroy callbacks (not for prototypes' destroy) ✔	§2.7
`@Profile`	core	register only when a profile expression matches (`"!dev"`, `"prod & eu"`)	§2.11
`@Conditional`	core	register only if a `Condition` matches	§2.11
`@EventListener`	core	receive application events (synchronous by default)	§2.12
`@Proxyable`	core (F7)	choose JDK or CGLIB proxy for one bean	§4.4
`@NullMarked` / `@Nullable`	`org.jspecify.annotations` (F7)	null-safety contract; `@Nullable` injection point = optional	§2.13
`@Inject`	`jakarta.inject` (optional jar)	standard alternative to `@Autowired`; `javax.inject` removed in F7	§2.4

Spring Boot and configuration

Annotation	Module / starter	Purpose	§
`@SpringBootApplication`	spring-boot-autoconfigure / `spring-boot-starter`	`@SpringBootConfiguration` + `@EnableAutoConfiguration` + `@ComponentScan`	§1.8
`@EnableAutoConfiguration`	spring-boot-autoconfigure	load `AutoConfiguration.imports` candidates and apply them after your beans	§3.10
`@AutoConfiguration`	spring-boot-autoconfigure	marks an auto-configuration class (`before` / `after` ordering)	§3.13
`@ConfigurationProperties("prefix")`	spring-boot	bind a group of properties to a typed object (record ideal)	§3.6
`@ConfigurationPropertiesScan` / `@EnableConfigurationProperties`	spring-boot	register `@ConfigurationProperties` classes	§3.6
`@DefaultValue`	spring-boot	default for a constructor-bound property ✔	§3.6
`@ConstructorBinding`	spring-boot	pick the binding constructor when there are several	§3.6
`@DurationUnit` / `@DataSizeUnit`	spring-boot	unit for plain numbers (default ms / bytes) ✔	§3.6
`@Validated` (on properties)	core + `spring-boot-starter-validation`	validate bound properties at startup	§3.6
`@ConditionalOnClass` / `OnMissingClass`	spring-boot-autoconfigure	class present / absent on the classpath	§3.10
`@ConditionalOnBean` / `OnMissingBean`	spring-boot-autoconfigure	bean present / absent (reliable in auto-configurations only)	§3.10
`@ConditionalOnProperty` / `OnBooleanProperty`	spring-boot-autoconfigure	property value / boolean switch (`OnBooleanProperty` since Boot 3.5)	§3.10
`@ConditionalOnWebApplication` / `OnThreading` / `OnCloudPlatform` / ...	spring-boot-autoconfigure	web type, virtual vs platform threads, platform	§3.10
`@PropertySource`	core	add a `.properties` file (low precedence, no YAML)	§3.9
`@EntityScan`	spring-boot-persistence (Boot 4 package)	where to find JPA entities outside the main package	§17.13

AOP and resilience

Annotation	Module / starter	Purpose	§
`@Aspect` (+ `@Component`)	aspectjweaver / `spring-boot-starter-aspectj` (Boot 4)	a class of pointcuts and advice; must also be a bean	§4.2
`@Pointcut`	aspectjweaver	named, reusable pointcut expression	§4.3
`@Before` / `@AfterReturning` / `@AfterThrowing` / `@After` / `@Around`	aspectjweaver	the five advice types; only `@Around` controls `proceed()`	§4.2
`@EnableAspectJAutoProxy`	core	switch on `@Aspect` support (Boot does it: `spring.aop.auto=true`)	§4.2
`@Retryable`	core `org.springframework.resilience` (F7)	retry on exception; `maxRetries` 3, `delay` 1000 ms, `multiplier` 1.0 ✔ — **needs `@EnableResilientMethods`**	§4.7
`@ConcurrencyLimit`	core (F7)	bulkhead: max concurrent calls, policy `BLOCK` or `REJECT` — needs `@EnableResilientMethods`	§4.7
`@EnableResilientMethods`	core (F7)	turns on `@Retryable` and `@ConcurrencyLimit` (no auto-configuration)	§4.7
`@CircuitBreaker` / `@Retry` / `@RateLimiter` / `@Bulkhead` / `@TimeLimiter`	`resilience4j-spring-boot4`	Resilience4j patterns with `fallbackMethod`	§16.8

Web: Spring MVC, WebFlux, REST

Annotation	Module / starter	Purpose	§
`@RequestMapping`	spring-web	class-level prefix / shared mapping attributes	§5.3
`@GetMapping` / `@PostMapping` / `@PutMapping` / `@PatchMapping` / `@DeleteMapping`	spring-web	map an HTTP method + path; `version = "1.1+"` (F7)	§5.3, §5.9
`@PathVariable`	spring-web	URL segment `{id}`	§5.4
`@RequestParam`	spring-web	query parameter / form field (`required = true` by default)	§5.4
`@RequestHeader` / `@CookieValue`	spring-web	header / cookie value	§5.4
`@RequestBody`	spring-web	body converted by an `HttpMessageConverter` (Jackson 3)	§5.4
`@RequestPart`	spring-web	part of a `multipart/form-data` request (`MultipartFile`)	§5.10
`@ModelAttribute`	spring-web	bind query params/form fields to an object	§5.4
`@ResponseStatus`	spring-web	fixed status for a handler or an exception class	§5.4, §5.7
`@ResponseBody`	spring-web	write the return value as the body	§5.3
`@ExceptionHandler`	spring-web	handle an exception type (in a controller or an advice)	§5.7
`@ControllerAdvice` / `@RestControllerAdvice`	spring-web	global exception handling (return `ProblemDetail`)	§5.7
`@Valid`	`jakarta.validation` + `-validation` starter	cascade validation into a body / nested object	§5.6
`@Validated`	core	validation **groups**; on a class: method validation via proxy	§5.6
`@CrossOrigin`	spring-web	per-controller CORS rules	§5.10
`@JsonProperty` / `@JsonIgnore`	`com.fasterxml.jackson.annotation`	JSON names / hidden fields (package unchanged in Jackson 3)	§5.5
`@JacksonComponent` / `@JacksonMixin`	spring-boot-jackson (Boot 4)	register (de)serializers / mix-ins (were `@JsonComponent` / `@JsonMixin`)	§5.5
`@EnableWebMvc`	spring-webmvc	full manual MVC control — **switches Boot's MVC auto-configuration off**	§3.12

HTTP clients

Annotation	Module / starter	Purpose	§
`@HttpExchange`	spring-web	HTTP service interface: shared URL prefix, headers, version	§6.6
`@GetExchange` / `@PostExchange` / `@PutExchange` / `@PatchExchange` / `@DeleteExchange`	spring-web	one HTTP call per interface method	§6.6
`@ImportHttpServices(group, types)`	spring-web (F7) + `spring-boot-starter-restclient`	register HTTP interface proxies per group, configured by `spring.http.serviceclient.<group>.*`	§6.6
`@ClientRegistrationId`	spring-security-oauth2-client	which OAuth2 registration's token to add (type-level since S7)	§11.10
`@LoadBalanced`	spring-cloud-commons	resolve `http://service-id` through Spring Cloud LoadBalancer	§16.5
`@FeignClient` / `@EnableFeignClients`	spring-cloud-openfeign (maintenance mode)	declarative client, Feign style	§16.6

Data access, transactions, JPA and Spring Data

Annotation	Module / starter	Purpose	§
`@Transactional`	spring-tx	declarative transaction (see §19.6) ✔	§7.8
`@EnableTransactionManagement`	spring-tx	switch on `@Transactional` (Boot does it); `rollbackOn = ALL_EXCEPTIONS` (F6.2)	§7.10
`@TransactionalEventListener`	spring-tx	run a listener at `AFTER_COMMIT` (default), `BEFORE_COMMIT`, `AFTER_ROLLBACK`, `AFTER_COMPLETION`	§7.15
`@Entity` / `@Table` / `@Column`	`jakarta.persistence` / `-data-jpa`	map a class / table / column	§8.3
`@Id` / `@GeneratedValue`	`jakarta.persistence`	primary key and strategy (`IDENTITY`, `SEQUENCE`, `UUID`, `AUTO`)	§8.3
`@ManyToOne` / `@OneToMany` / `@OneToOne` / `@ManyToMany`	`jakarta.persistence`	relationships; to-one EAGER, to-many LAZY by default	§8.5
`@JoinColumn` / `mappedBy`	`jakarta.persistence`	owning side (FK) / inverse side	§8.5
`@Enumerated(EnumType.STRING)` / `@Embedded` / `@Embeddable` / `@Transient`	`jakarta.persistence`	enum as text / value object / not stored	§8.3
`@Version`	`jakarta.persistence` (JPA) or `org.springframework.data.annotation` (Data JDBC, Mongo)	optimistic locking column	§8.14, §9.2
`@Query`	spring-data-jpa / -jdbc / ...	JPQL (JPA), SQL (JDBC), `nativeQuery = true`	§8.10
`@Modifying`	spring-data-jpa / -jdbc	UPDATE/DELETE query; `clearAutomatically`	§8.10
`@Param`	spring-data-commons	name a query parameter (optional with `-parameters`)	§8.10
`@EntityGraph(attributePaths = ...)`	spring-data-jpa	fetch associations in the same query (left join)	§8.7
`@Lock(LockModeType.PESSIMISTIC_WRITE)`	spring-data-jpa	`SELECT ... FOR UPDATE`	§8.14
`@BatchSize`	Hibernate	batch-load lazy collections (N+1 fix)	§8.7
`@DynamicUpdate`	Hibernate	UPDATE only changed columns	§8.4
`@EnableJpaAuditing` / `@EnableJdbcAuditing`	spring-data-jpa / -jdbc	switch on auditing	§8.13, §9.2
`@CreatedDate` / `@LastModifiedDate` / `@CreatedBy` / `@LastModifiedBy`	spring-data-commons	audit columns	§8.13
`@MappedCollection(idColumn, keyColumn)`	spring-data-relational	child collection of a Data JDBC aggregate	§9.2
`@RepositoryRestResource`	spring-data-rest	export (or hide) a repository as HAL endpoints	§9.3
`@Document` / `@Indexed`	spring-data-mongodb	MongoDB document / index	§9.5
`@RedisHash(timeToLive = ...)`	spring-data-redis	Redis repository entity with TTL	§9.6

Caching

Annotation	Module / starter	Purpose	§
`@EnableCaching`	core / `spring-boot-starter-cache`	switch on the cache proxy — without it nothing is cached	§9.8
`@Cacheable`	core	return the cached value; run the method only on a miss (`sync`, `condition`, `unless`)	§9.8
`@CachePut`	core	always run, then store the result	§9.8
`@CacheEvict`	core	remove a key (`allEntries = true` to clear)	§9.8
`@Caching` / `@CacheConfig`	core	several cache ops on one method / class defaults	§9.8

Security

Annotation	Module / starter	Purpose	§
`@EnableWebSecurity`	spring-security-config	web security (Boot applies it for you)	§10.3
`@EnableMethodSecurity`	spring-security-config	switch on method security annotations	§10.7
`@PreAuthorize` / `@PostAuthorize`	spring-security-core	SpEL rule before / after the call (`returnObject`)	§10.7
`@PreFilter` / `@PostFilter`	spring-security-core	filter a collection argument / result	§10.7
`@Secured` / `@RolesAllowed`	spring-security-core / `jakarta.annotation.security`	legacy role checks (`securedEnabled`, `jsr250Enabled`)	§10.7
`@AuthenticationPrincipal`	spring-security-core	inject the principal (`UserDetails`, `Jwt`, `OidcUser`)	§10.4
`@EnableMultiFactorAuthentication`	spring-security-config (S7)	require several `FACTOR_*` authorities	§10.4

Messaging, async and scheduling

Annotation	Module / starter	Purpose	§
`@EnableAsync`	core	switch on `@Async` (Boot configures the executor but does **not** enable it)	§14.4
`@Async("executor")`	core	run on an executor; return `void` or `CompletableFuture`	§14.4
`@EnableScheduling`	core	switch on `@Scheduled`	§14.7
`@Scheduled(fixedRate / fixedDelay / cron)`	core	timer jobs; cron has **six** fields ✔	§14.7
`@KafkaListener(topics, groupId, concurrency)`	spring-kafka / `spring-boot-starter-kafka`	Kafka consumer method	§14.12
`@RetryableTopic` / `@DltHandler`	spring-kafka	non-blocking retries via retry topics; dead-letter handler	§14.15
`@RabbitListener`	spring-rabbit / `spring-boot-starter-amqp`	RabbitMQ consumer method	§14.18
`@JmsListener`	spring-jms	JMS consumer method	§14.18
`@SchedulerLock`	ShedLock (third party)	run a `@Scheduled` job on one instance only	§14.7

Observability, cloud and AI

Annotation	Module / starter	Purpose	§
`@Observed`	micrometer-observation (+ AspectJ starter)	timer + span around a method; needs `management.observations.annotations.enabled=true`	§15.6
`@Timed` / `@Counted` / `@NewSpan`	Micrometer	metric / span annotations (same switch as `@Observed`)	§15.6
`@EnableConfigServer`	spring-cloud-config-server	make the app a Config Server	§16.3
`@RefreshScope`	spring-cloud-context	re-create the bean after a config refresh	§16.3
`@EnableEurekaServer`	spring-cloud-netflix-eureka-server	run a Eureka registry	§16.4
`@Tool` / `@ToolParam`	spring-ai-model (Spring AI 2.0)	expose a method to the model for tool calling	§17.8
`@McpTool` / `@McpToolParam`	Spring AI 2.0 MCP (`org.springframework.ai.mcp.annotation`)	expose a tool from an MCP server	§17.10

Testing annotations have their own sheet: §19.9.

19.4 Starters in Boot 4.1

A **starter** is a code-free dependency that brings one capability's libraries plus the Boot module holding its auto-configuration (§1.7). Boot 4 made them **modular**: one starter per technology, and a matching `-test` starter that brings the test slice (a test that starts only one layer of the app, such as the web layer) for it and already includes `spring-boot-starter-test` (§3.14, §12.3). All names below are managed by the 4.1.1 BOM — never add a `<version>`.

The starters you will use most

Capability	Starter	Test starter (slice, test clients)	§
Core only (no web)	`spring-boot-starter`	`spring-boot-starter-test`	§1.7
Servlet web / REST (Spring MVC + Tomcat + Jackson 3)	`spring-boot-starter-webmvc`	`spring-boot-starter-webmvc-test` (`@WebMvcTest`, MockMvc, `RestTestClient`)	§5.2
Reactive web (WebFlux + Reactor Netty)	`spring-boot-starter-webflux`	`spring-boot-starter-webflux-test` (`@WebFluxTest`, `WebTestClient`)	§13.10
Blocking HTTP client (`RestClient.Builder`, HTTP service groups)	`spring-boot-starter-restclient`	`spring-boot-starter-restclient-test` (`@RestClientTest`)	§6.1
Reactive HTTP client (`WebClient.Builder`)	`spring-boot-starter-webclient`	`spring-boot-starter-webclient-test`	§13.13
JSON (Jackson 3)	`spring-boot-starter-jackson`	`spring-boot-starter-jackson-test` (`@JsonTest`)	§5.5
Bean Validation (Hibernate Validator)	`spring-boot-starter-validation`	`spring-boot-starter-validation-test`	§5.6
AOP with `@Aspect`	`spring-boot-starter-aspectj`	`spring-boot-starter-aspectj-test`	§4.2
JDBC (`DataSource`, HikariCP, `JdbcClient`)	`spring-boot-starter-jdbc`	`spring-boot-starter-jdbc-test` (`@JdbcTest`)	§7.2
JPA (Hibernate + Spring Data JPA)	`spring-boot-starter-data-jpa`	`spring-boot-starter-data-jpa-test` (`@DataJpaTest`)	§8.2
Spring Data JDBC	`spring-boot-starter-data-jdbc`	`spring-boot-starter-data-jdbc-test` (`@DataJdbcTest`)	§9.2
R2DBC	`spring-boot-starter-data-r2dbc`	`spring-boot-starter-data-r2dbc-test` (`@DataR2dbcTest`)	§13.14
MongoDB / Redis	`spring-boot-starter-data-mongodb` / `-data-redis`	`...-data-mongodb-test` / `...-data-redis-test`	§9.5, §9.6
Spring Data REST	`spring-boot-starter-data-rest`	`spring-boot-starter-data-rest-test`	§9.3
Schema migrations	`spring-boot-starter-flyway` / `-liquibase`	`...-flyway-test` / `...-liquibase-test`	§8.16, §9.4
Cache abstraction	`spring-boot-starter-cache` (+ `caffeine`)	`spring-boot-starter-cache-test`	§9.8
Security	`spring-boot-starter-security`	`spring-boot-starter-security-test`	§10.1
OAuth2 resource server / client / authorization server	`spring-boot-starter-security-oauth2-resource-server` / `-client` / `-authorization-server`	`...-test` for each	§11.3, §11.9, §11.14
Kafka	`spring-boot-starter-kafka` (+ `-jackson` for JSON)	`spring-boot-starter-kafka-test` (`@EmbeddedKafka`)	§14.10
RabbitMQ	`spring-boot-starter-amqp`	`spring-boot-starter-amqp-test`	§14.18
Actuator	`spring-boot-starter-actuator`	`spring-boot-starter-actuator-test`	§15.2
Tracing + OTLP export	`spring-boot-starter-opentelemetry` (or `-zipkin` for Brave)	`spring-boot-starter-opentelemetry-test` (`@AutoConfigureTracing`)	§15.7
Metrics only	`spring-boot-starter-micrometer-metrics` (+ a registry, e.g. `micrometer-registry-prometheus`)	`...-micrometer-metrics-test` (`@AutoConfigureMetrics`)	§15.5
HATEOAS	`spring-boot-starter-hateoas`	`...-hateoas-test`	§5.12
Testcontainers support	`spring-boot-testcontainers` (module, test scope)	—	§12.14
Also available	`-websocket`, `-graphql`, `-grpc-server` / `-grpc-client` (Boot 4.1), `-batch`, `-batch-jdbc`, `-integration`, `-quartz`, `-mail`, `-jms`, `-artemis`, `-pulsar`, `-thymeleaf`, `-session-data-redis`, `-session-jdbc`, `-jooq`, `-jetty`	most have a `-test` companion	§1.3

Renamed, deprecated and removed starters

Boot 3 name	Boot 4.1 status	Use instead	§
`spring-boot-starter-web`	**deprecated** alias (still in the BOM)	`spring-boot-starter-webmvc`	§1.7
`spring-boot-starter-web-services`	**deprecated** alias	`spring-boot-starter-webservices`	§17.13
`spring-boot-starter-oauth2-resource-server` / `-oauth2-client` / `-oauth2-authorization-server`	**deprecated** aliases	`spring-boot-starter-security-oauth2-...`	§11.3
`spring-boot-starter-aop`	**removed**	`spring-boot-starter-aspectj`	§4.2
`spring-boot-starter-undertow`	**removed** (Undertow lacks Servlet 6.1)	Tomcat (default) or `spring-boot-starter-jetty`	§17.13
`spring-boot-starter-json`	still available	`spring-boot-starter-jackson` is the Boot 4 name	§1.7
`flyway-core` / `liquibase-core` alone	no longer auto-configured	`spring-boot-starter-flyway` / `-liquibase`	§8.16
`spring-kafka` alone	still works as a library, but use the starter	`spring-boot-starter-kafka`	§14.10
part of `-web` / `-webflux` in Boot 3	split out	`spring-boot-starter-restclient`, `spring-boot-starter-webclient`	§6.1, §13.10
`spring-boot-starter-batch` (with JDBC metadata)	batch now runs without a database by default	`spring-boot-starter-batch-jdbc`	§17.13
(migration aid)	new	`spring-boot-starter-classic`, `spring-boot-starter-test-classic`: the old "everything" layout while you migrate	§3.14

Third-party starters are named `<name>-spring-boot-starter` (`mybatis-spring-boot-starter`); the `spring-boot-*` prefix is reserved (§3.13).

19.5 Key properties cheat sheet

Defaults are the Boot 4.1.1 metadata values; "—" means *no default: the feature is off, or the value comes from somewhere else*. Precedence (which source wins) and relaxed binding (several spellings of one property name, e.g. `SHOP_NAME` for `shop.name`, are accepted) are at the end of this section.

Core, profiles and startup

Property	Default	What it does	§
`spring.application.name`	—	name in logs, traces, Config Server, metrics	§1.9
`spring.profiles.active` / `.default` / `.include`	— / `default` / —	activate profiles (not allowed inside profile-specific files)	§3.7
`spring.profiles.group.<name>`	—	one name activates several profiles	§3.7
`spring.config.import`	—	extra files, `optional:`, `configtree:`, `configserver:`	§3.8
`spring.config.activate.on-profile`	—	make a document of a multi-document file conditional	§3.2
`spring.autoconfigure.exclude`	—	switch one auto-configuration off	§3.12
`debug`	`false`	print the condition evaluation report	§3.11
`spring.main.lazy-initialization`	`false`	make every bean lazy	§2.10
`spring.main.allow-circular-references`	`false`	Boot rejects field/setter cycles (since 2.6)	§2.9
`spring.main.allow-bean-definition-overriding`	`false`	two beans with one name → `BeanDefinitionOverrideException`	§2.3
`spring.main.web-application-type`	detected	`servlet`, `reactive`, `none` (MVC wins if both starters present)	§13.10
`spring.main.keep-alive`	`false`	keep the JVM alive when only virtual (daemon) threads run	§14.5
`spring.aop.proxy-target-class`	`true`	Boot uses CGLIB proxies for everything	§4.4
`spring.aop.auto`	`true`	`@EnableAspectJAutoProxy` applied by Boot	§4.2
`spring.lifecycle.timeout-per-shutdown-phase`	`30s` ✔	how long graceful shutdown waits	§15.9
`spring.devtools.restart.enabled`	`true`	DevTools restart (off for `java -jar` and tests)	§1.10
`spring.devtools.livereload.enabled`	`false`	LiveReload — off in Boot 4.0, deprecated in 4.1	§1.10

Web server, MVC and JSON

Property	Default	What it does	§
`server.port`	`8080`	HTTP port; `0` = random free port	§1.9
`server.shutdown`	`graceful`	`immediate` turns graceful shutdown off (default since Boot 3.4)	§15.9
`server.servlet.context-path`	–	URL prefix of the whole app	§5.2
`server.forward-headers-strategy`	–	`framework` honours `X-Forwarded-*` behind a proxy	§5.4
`server.servlet.session.timeout`	`30m`	HTTP session timeout	§10.10
`server.servlet.session.cookie.secure` / `.http-only` / `.same-site`	–	harden the session cookie	§10.10
`spring.web.error.include-stacktrace`	`never`	stack traces in Boot's error body (Boot 4 name; `server.error.*` no longer bound)	§15.12
`spring.mvc.problemdetails.enabled`	`false`	RFC 9457 bodies for MVC exceptions without your own advice	§5.7
`spring.webflux.problemdetails.enabled`	`false`	the same for WebFlux	§13.10
`spring.mvc.contentnegotiation.favor-parameter`	`false`	allow `?format=csv`	§5.8
`spring.servlet.multipart.max-file-size` / `max-request-size`	`1MB` / `10MB`	upload limits (413 when exceeded)	§5.10
`spring.jackson.default-property-inclusion`	–	e.g. `non_null`	§5.5
`spring.jackson.serialization.*` / `deserialization.*`	–	Jackson feature switches (`indent-output`...)	§5.5
`spring.jackson.json.read.*` / `spring.jackson.json.write.*`	–	JSON parser/generator features (Boot 4; were `spring.jackson.parser/generator.*`)	§5.5
`spring.jackson.use-jackson2-defaults`	`false`	restore Jackson 2 defaults while migrating	§17.13
`spring.threads.virtual.enabled`	`false`	virtual threads for Tomcat, `@Async`, `@Scheduled`, listeners (Java 21+) ✔	§15.10

Jackson 3 defaults under Boot 4 (✔ proved): `LocalDate` written as `"2026-09-28"`, unknown JSON properties ignored, `JacksonException` is unchecked (a `RuntimeException`, so callers need not catch it) (§5.5).

API versioning (F7 / Boot 4)

Property	Meaning	§
`spring.mvc.apiversion.use.header`	read the version from this header (e.g. `X-API-Version`)	§5.9
`spring.mvc.apiversion.use.query-parameter` / `.use.path-segment` / `.use.media-type-parameter`	other places to read it	§5.9
`spring.mvc.apiversion.default`	version assumed when the request has none	§5.9
`spring.mvc.apiversion.required`	reject requests without a version (400)	§5.9
`spring.mvc.apiversion.supported` / `.detect-supported`	extra supported versions / detect them from mappings	§5.9
`spring.webflux.apiversion.*`	the same for WebFlux	§17.13
`spring.http.serviceclient.<group>.apiversion.insert.header`	client side: send the version in this header	§6.7

HTTP clients

Property	Default	What it does	§
`spring.http.clients.connect-timeout` / `read-timeout`	– (library default)	global timeouts for auto-configured clients (Boot 4 name; `spring.http.client.*` deprecated)	§6.4
`spring.http.clients.imperative.factory`	detected	`http-components`, `jetty`, `reactor`, `jdk`, `simple`	§6.4
`spring.http.clients.reactive.connector`	detected	the same for `WebClient`	§6.4
`spring.http.clients.redirects`	–	`follow`, `dont-follow`, `follow-when-possible`	§6.4
`spring.http.serviceclient.<group>.base-url`	–	base URL of an `@ImportHttpServices` group (none + Spring Cloud = load-balanced)	§6.6, §16.6
`spring.http.serviceclient.<group>.read-timeout` / `.default-header.<name>`	–	per-group timeout / header	§6.6

DataSource, JDBC, transactions

Property	Default	What it does	§
`spring.datasource.url` / `.username` / `.password`	–	connection; no URL + embedded DB on the classpath → in-memory DB ✔	§7.2
`spring.datasource.generate-unique-name`	`true`	random in-memory DB name per context ✔	§7.2
`spring.datasource.hikari.maximum-pool-size`	`10` (HikariCP) ✔	most connections	§7.2
`spring.datasource.hikari.minimum-idle`	= max	leave unset: fixed-size pool	§7.2
`spring.datasource.hikari.connection-timeout`	`30000` ms ✔	wait for a free connection, then fail	§7.2
`spring.datasource.hikari.max-lifetime` / `idle-timeout` / `keepalive-time`	30 min / 10 min / 2 min	connection recycling	§7.2
`spring.datasource.hikari.leak-detection-threshold`	`0` (off)	log connections held too long	§7.2
`spring.datasource.connection-fetch`	`eager`	`lazy` = no physical connection until the first statement (Boot 4.1)	§7.2
`spring.sql.init.mode`	`embedded`	run `schema.sql` / `data.sql`: `embedded`, `always`, `never`	§7.4
`spring.transaction.default-timeout`	–	default transaction timeout	§7.12
`spring.persistence.exceptiontranslation.enabled`	`true`	`@Repository` exception translation (was `spring.dao.*`)	§7.5

JPA, Hibernate, migrations

Property	Default	What it does	§
`spring.jpa.hibernate.ddl-auto`	`create-drop` for an embedded DB without a migration tool, else `none`	`validate` in production	§8.2
`spring.jpa.open-in-view`	`true` (+ startup warning)	set `false` for APIs	§8.15
`spring.jpa.show-sql`	`false`	print SQL (`logging.level.org.hibernate.SQL=DEBUG` is better)	§8.2
`spring.jpa.defer-datasource-initialization`	`false`	run `data.sql` after Hibernate created the tables	§7.4
`spring.jpa.properties.hibernate.default_batch_fetch_size`	–	N+1 safety net (e.g. `16`)	§8.7
`spring.jpa.properties.hibernate.jdbc.batch_size` / `order_inserts`	–	batched inserts (needs SEQUENCE ids)	§8.18
`spring.jpa.properties.*`	–	passed to Hibernate unchanged	§8.2
`spring.flyway.enabled` / `.locations`	`true` / `classpath:db/migration`	Flyway (needs `spring-boot-starter-flyway`)	§9.4
`spring.flyway.baseline-on-migrate` / `.clean-disabled`	`false` / `true`	adopt an existing DB / keep `clean` disabled	§9.4
`spring.liquibase.change-log`	`classpath:/db/changelog/db.changelog-master.yaml`	Liquibase master changelog	§9.4
`spring.data.rest.base-path` / `.detection-strategy`	– / `default`	Data REST prefix / `annotated` = only `@RepositoryRestResource`	§9.3
`spring.mongodb.uri`	`mongodb://localhost/test`	MongoDB connection (Boot 4; `spring.data.mongodb.uri` is no longer bound)	§9.5
`spring.data.redis.host` / `.port`	`localhost` / `6379`	Redis connection	§9.6
`spring.r2dbc.url`	–	reactive SQL connection	§13.14

Cache

Property	Default	What it does	§
`spring.cache.type`	detected (Redis before Caffeine before Simple)	force a provider, or `none` in tests	§9.8
`spring.cache.cache-names`	—	fixed set of caches (typos fail)	§9.8
`spring.cache.caffeine.spec`	—	e.g. `maximumSize=500,expireAfterWrite=10m`	§9.8
`spring.cache.redis.time-to-live`	— (no expiry)	TTL for Redis caches	§9.8
`spring.cache.redis.cache-null-values` / `.key-prefix`	`true` / —	cache `null`s / key prefix (version it)	§9.8

Security

Property	Default	What it does	§
`spring.security.user.name` / `.password` / `.roles`	`user` / random UUID (logged) / —	the dev-only generated user	§10.1
`spring.security.oauth2.resourceserver.jwt.issuer-uri`	—	discover JWKS, check `iss`	§11.3
`spring.security.oauth2.resourceserver.jwt.jwk-set-uri` / `.public-key-location`	—	keys without discovery	§11.3
`spring.security.oauth2.resourceserver.jwt.audiences`	—	check `aud` (do it!)	§11.3
`spring.security.oauth2.resourceserver.jwt.authorities-claim-name` / `.authority-prefix` / `.principal-claim-name`	— (`scope`/`scp` → `SCOPE_`)	map roles from a claim without code	§11.5
`spring.security.oauth2.resourceserver.opaquetoken.introspection-uri` / `.client-id` / `.client-secret`	—	opaque tokens	§11.7
`spring.security.oauth2.client.registration.<id>.*` / `.provider.<id>.issuer-uri`	—	OAuth2 login / client credentials	§11.9, §11.10

Actuator, health, metrics, tracing, logging

Property	Default	What it does	§
`management.endpoints.web.exposure.include` / `.exclude`	`health`	which endpoints are reachable over HTTP (`*` = all)	§15.2
`management.endpoints.web.base-path`	`/actuator`	URL prefix	§15.2
`management.endpoints.access.default` / `management.endpoint.<id>.access`	per endpoint (`shutdown` = `none`)	`none`, `read-only`, `unrestricted` (since 3.4)	§15.2
`management.server.port`	— (same port)	separate management port	§15.2
`management.endpoint.health.show-details`	`never`	`when-authorized`, `always`	§15.3
`management.endpoint.health.probes.enabled`	`true` (Boot 4)	`/actuator/health/liveness` and `/readiness`	§15.3
`management.endpoint.health.group.<name>.include`	—	custom health group	§15.3
`management.endpoint.env.show-values`	`never`	mask values in `env`/`configprops`	§15.13
`management.info.env.enabled` / `.java.enabled`	`false` / `false`	info contributors	§15.4
`management.metrics.tags.<key>`	—	common tag on every meter	§15.5
`management.metrics.distribution.percentiles-histogram.<meter>`	—	histogram buckets for p95/p99	§15.5
`management.observations.annotations.enabled`	`false`	activate `@Observed`, `@Timed`, `@Counted`, `@NewSpan`	§15.6
`management.observations.enable.<prefix>`	—	switch observation families off	§15.6
`management.tracing.sampling.probability`	`0.1`	share of traces recorded	§15.7
`management.tracing.export.enabled`	`true`	export spans (was `management.tracing.enabled`)	§17.13
`management.opentelemetry.tracing.export.otlp.endpoint`	—	OTLP collector URL (was `management.otlp.tracing.endpoint`)	§15.7
`management.tracing.propagation.type`	W3C	`b3` for legacy Zipkin services	§16.9
`management.tracing.baggage.remote-fields` / `.correlation.fields`	—	baggage in headers / in the MDC	§15.7
`logging.level.<logger>`	`INFO` root	per-logger level (`POST /actuator/loggers/...` at runtime)	§15.2
`logging.structured.format.console` / `.file`	—	`ecs`, `logstash`, `gelf` JSON logs (Boot 3.4+)	§15.8
`logging.pattern.correlation`	Boot's `[traceId-spanId]`	how trace ids appear in text logs	§15.7

Threads, async, scheduling, Kafka, RabbitMQ

Property	Default	What it does	§
`spring.task.execution.pool.core-size`	`8` ✔	`applicationTaskExecutor` threads (`@Async`)	§14.4
`spring.task.execution.pool.queue-capacity` / `.max-size`	unbounded ✔ / unlimited	bound the queue!	§14.4
`spring.task.execution.thread-name-prefix`	`task-` ✔	thread names	§14.4
`spring.task.execution.mode`	`auto`	`force` keeps Boot's executor next to your own `Executor` beans	§14.4
`spring.task.execution.propagate-context`	`false`	copy tracing/MDC context into `@Async` threads	§14.4
`spring.task.execution.simple.concurrency-limit`	–	cap for the virtual-thread executor	§14.5
`spring.task.execution.shutdown.await-termination`	`false`	finish queued tasks on shutdown	§14.4
`spring.task.scheduling.pool.size`	`1` ✔	scheduler threads for all `@Scheduled` methods	§14.7
`spring.reactor.context-propagation`	`limited`	`auto` bridges ThreadLocals and the Reactor `Context`	§13.9
`spring.kafka.bootstrap-servers`	`localhost:9092`	brokers (tests: `${spring.embedded.kafka.brokers}`)	§14.10
`spring.kafka.consumer.group-id` / `.auto-offset-reset`	– / Kafka's `latest`	consumer group / where a new group starts	§14.10
`spring.kafka.consumer.isolation-level`	`read-uncommitted`	`read-committed` to hide aborted transactions	§14.16
`spring.kafka.listener.ack-mode` / `.concurrency`	`BATCH` (container default) / 1	offset commits / consumer threads	§14.12
`spring.kafka.producer.acks` / `.transaction-id-prefix`	Kafka's `all` / –	durability / transactional template	§14.16
`spring.kafka.template.observation-enabled` / `listener.observation-enabled`	`false` / `false`	tracing through Kafka headers	§16.9
`spring.rabbitmq.host` / `.listener.simple.prefetch`	`localhost` / –	broker / unacknowledged messages per consumer	§14.18
`spring.rabbitmq.listener.simple.retry.max-retries`	`3`	retries **after** the first attempt (Boot 4; replaces `max-attempts`)	§14.18

Precedence and relaxed binding (✔ proved)

Highest wins (§3.3, §12.15):

1. In tests: `@DynamicPropertySource` > `@TestPropertySource` > `@SpringBootTest(properties = ...)`.
2. Command-line arguments `--shop.name=...`
3. `SPRING_APPLICATION_JSON`
4. Java system properties `-Dshop.name=...`
5. OS environment variables `SHOP_NAME=...`
6. `random.*`
7. Profile-specific files `application-{profile}.*` (outside the jar > inside)
8. `application.properties` / `.yaml` (outside the jar > inside; an imported file > its importer)
9. `@PropertySource` files
10. `SpringApplication.setDefaultProperties(...)`

Rule	Example	§
env var name = upper case, `.` → `_`, drop `-`	`SHOP_CHECKOUTTIMEOUT=2m` → `shop.checkout-timeout` ✔	§3.4
list index	`SHOP_CURRENCIES_0=USD` → `shop.currencies[0]` ✔	§3.4
legacy form (`-` → `_`) is also accepted by the binder	`SHOP_CHECKOUT_TIMEOUT=3m` also binds ✔ – prefer the documented form	§3.4
`Duration`	`45s`, `5m`, `PT1H`; plain `500` = milliseconds ✔	§3.6
`DataSize`	`10MB` = 10 485 760 bytes; plain `1024` = bytes ✔	§3.6
`@Value`	`${key}` fails if missing; `${key:default}`; `#{${key:3} + 1}` = SpEL ✔	§3.5

19.6 Transactions: `@Transactional` at a glance

A Spring transaction = a **proxy** + a `PlatformTransactionManager` + a connection **bound to the current thread** (stored per thread, so every data-access call on that thread reuses the same connection) (§7.7). Everything below applies only to calls that go **through the proxy** (§7.13).

Attributes and their defaults (✔ proved)

Attribute	Default	Meaning	§
`propagation`	`REQUIRED`	what to do when a transaction already exists (table below)	§7.9
`isolation`	`DEFAULT` = the database's default	only applied when this method **starts** the transaction	§7.11
`readOnly`	`false`	JDBC: read-only connection (PostgreSQL rejects writes); JPA: no dirty checking/flush; can route to a replica	§7.12
`timeout`	`-1` (none)	seconds; enforced on the next SQL statement	§7.12
`rollbackFor` / `noRollbackFor`	–	add / remove exception types from the rollback rule	§7.10
`transactionManager` (`value`)	the primary manager	pick one of several managers	§7.8
`label`	–	free-form labels for custom managers	§7.8

Where: on the **service method** that is one use case. Not on controllers (too wide), not only on repositories (too narrow). A class-level `@Transactional(readOnly = true)` + method-level `@Transactional` on writes is the Spring Data pattern (§7.8).

Propagation (✔ all seven proved)

Propagation	A transaction exists	None exists	Typical use
`REQUIRED` (default)	**join** it (same connection)	start one	almost everything
`REQUIRES_NEW`	**suspend** it, start an independent one on a **second connection**; commits even if the caller later rolls back	start one	audit log that must survive a rollback
`NESTED`	**savepoint** inside it; an inner failure rolls back to the savepoint only	start one	"try this part" (JDBC; avoid with JPA)
`MANDATORY`	join it	**throw** `IllegalTransactionStateException`	must be called inside a transaction
`SUPPORTS`	join it	run **without** a transaction	read helpers that don't care
`NOT_SUPPORTED`	**suspend** it, run without	run without	slow non-transactional work inside a flow
`NEVER`	**throw** `IllegalTransactionStateException`	run without	guard: never inside a transaction

The trap ✔: an inner `REQUIRED` method throws, the outer method **catches** the exception and returns normally → the shared transaction was marked **rollback-only** → Spring rolls back at commit and throws `UnexpectedRollbackException` (§7.9).

Rollback rules (✔ proved)

Thrown out of the method	Default	`rollbackFor = X`	`noRollbackFor = X`	`rollbackOn = ALL_EXCEPTIONS` (F6.2)
`RuntimeException` / `Error`	**rollback**	rollback	commit if it is an X	rollback
checked `Exception`	**commit** (!)	rollback if it is an X	commit	rollback
caught **inside** the method	commit – the proxy never sees it	commit	commit	commit

Global switch: `@EnableTransactionManagement(proxyTargetClass = true, rollbackOn = RollbackOn.ALL_EXCEPTIONS)` (§7.10). To mark a transaction without throwing: `TransactionAspectSupport.currentTransactionStatus().setRollbackOnly()` or `status.setRollbackOnly()` in a `TransactionTemplate`.

Isolation levels

Isolation	Prevents	Notes
`DEFAULT`	whatever the database default prevents	PostgreSQL, Oracle, SQL Server: READ COMMITTED; MySQL InnoDB: REPEATABLE READ
`READ_UNCOMMITTED`	nothing guaranteed (dirty reads possible)	PostgreSQL treats it as READ COMMITTED
`READ_COMMITTED`	dirty reads	non-repeatable reads and lost updates still possible (proved in §7.11)
`REPEATABLE_READ`	+ non-repeatable reads	PostgreSQL: one snapshot; concurrent update → serialization failure (`40001`) → retry the whole transaction
`SERIALIZABLE`	+ phantoms, write skew	expect serialization failures; always retry

The anomalies themselves (wrong results that concurrent transactions can produce, such as dirty reads, non-repeatable reads and lost updates): SQL 13.4.

Transaction managers and helpers

You use	Manager Boot creates	§
JDBC (`spring-boot-starter-jdbc`)	`JdbcTransactionManager` (a `DataSourceTransactionManager`)	§7.7
JPA (`spring-boot-starter-data-jpa`)	`JpaTransactionManager` (JDBC on the same `DataSource` joins it)	§7.7
R2DBC	`R2dbcTransactionManager` (context-bound, not thread-bound)	§13.14
Kafka with `spring.kafka.producer.transaction-id-prefix`	`KafkaTransactionManager`	§14.16
Programmatic	`TransactionTemplate.execute(status -> ...)` — no proxy, no self-invocation problem	§7.14
After-commit side effects	`@TransactionalEventListener` (`AFTER_COMMIT` default; skipped without a transaction unless `fallbackExecution = true` ; a transactional listener must use `REQUIRES_NEW`)	§7.15

"Why is my `@Transactional` not working?"

Symptom	Cause	§
partial writes after an exception	self-invocation (`this.save()`), `private` method, bean made with `new`	§7.13, §4.5
`NullPointerException` inside a `final` `@Transactional` method	the method ran on the CGLIB proxy instance, whose fields are `null`	§4.5
data committed although the method threw	checked exception without `rollbackFor` , or exception caught inside	§7.10
`UnexpectedRollbackException`	inner `REQUIRED` failure was caught	§7.9
`isolation` / `timeout` ignored	the method joined an existing transaction	§7.11
works in tests, fails in production	the test was `@Transactional` and supplied the transaction	§12.7
`@Async` code doesn't see uncommitted data	transactions are bound to one thread	§7.13
pool exhausted under load	`REQUIRES_NEW` holds two connections; remote calls inside transactions	§7.9, §7.13

19.7 Bean scopes and the lifecycle

Scopes (✔ singleton and prototype proved)

Scope	Instances	Destroy callbacks?	Declare with	§
`singleton` (default)	one per container, shared by all threads — keep it **stateless**	yes	nothing	§2.6
`prototype`	a new one per `getBean`/injection	**no** — Spring forgets it	`@Scope("prototype")`	§2.6
`request`	one per HTTP request	yes	`@RequestScope` (scoped proxy)	§2.6
`session`	one per HTTP session	yes	`@SessionScope`	§2.6
`application`	one per `ServletContext`	yes	`@ApplicationScope`	§2.6
`websocket`	one per WebSocket session	yes	`@Scope("websocket")`	§2.6
refresh (Spring Cloud)	re-created after a config refresh	yes	`@RefreshScope`	§16.3

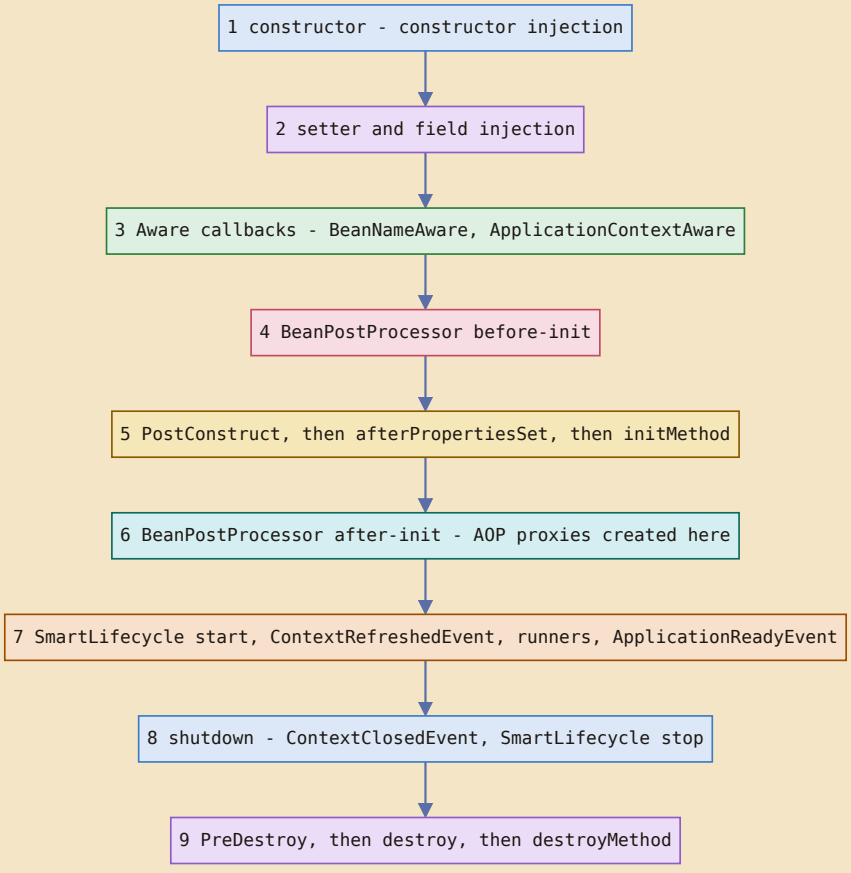
A prototype injected into a singleton is created **once** and stays ✔. For a fresh one each time: `ObjectProvider<T>.getObject()`, a `@Lookup` method, or a scoped proxy (a stand-in that looks up the right instance on every call) (§2.6).

How Spring picks one bean among several (✔ proved)

1. `@Qualifier("name")` on the injection point
2. `@Primary` bean
3. the only non-`@Fallback` bean (F6.2)
4. a bean whose name equals the parameter name (needs `-parameters` — a compiler flag that keeps parameter names in the bytecode — which the Boot parent sets)
5. otherwise `NoUniqueBeanDefinitionException`

`List<T>` gets all beans ordered by `@Order` (lowest first); `Map<String, T>` is keyed by bean name. Optional dependencies: `Optional<T>`, `ObjectProvider<T>`, `@Nullable`, `@Autowired(required = false)` (§2.5).

Lifecycle order (✔ proved)



Proved order: init `[constructor, @PostConstruct, afterPropertiesSet, initMethod]`, destroy `[@PreDestroy, destroy(), destroyMethod]`, and no `@PreDestroy` for the prototype.

Hook	Use for	§
`@PostConstruct` / `@PreDestroy`	default choice: validate, warm up, release	§2.7
`InitializingBean` / `DisposableBean`	framework code (couples you to Spring)	§2.7
`@Bean(initMethod, destroyMethod)`	third-party classes; a public `close()` / `shutdown()` is inferred	§2.7
`SmartLifecycle` (`getPhase()`)	things that *run*: pollers, listeners (start low phase first, stop it last)	§2.7
`CommandLineRunner` / `ApplicationRunner`	one-off startup work with the arguments	§1.8
`@EventListener(ApplicationReadyEvent.class)`	"after the app is fully started"	§2.7
`BeanFactoryPostProcessor` (declare `static`)	change bean **definitions** before any bean exists	§2.8
`BeanPostProcessor` (declare `static`)	change or wrap every bean **instance**	§2.8

Pitfall: a `@Transactional` call from `@PostConstruct` has no transaction — the proxy is created in step 6, after step 5, and the call is on `this` anyway (§2.7).

19.8 Spring Security 7 DSL snippets

Security 7 accepts only the **lambda DSL** (domain-specific language: configuration written as method calls that take lambdas, i.e. short inline functions `x -> ...`): one lambda per feature, no `.and()`, `authorizeHttpRequests` (not `authorizeRequests`), path matchers built on `PathPattern` (§10.3). The four snippets below are compiled by every build of `SecuritySnippets.java`, so each DSL call exists in Security 7.1.1. Rules are checked **top to bottom, first match wins** (§10.6).

Form login + HTTP Basic (browser app)

```
@Bean
SecurityFilterChain web(HttpSecurity http) throws Exception {
    http.authorizeHttpRequests(auth -> auth                      // first matching rule wins
            .requestMatchers("/", "/public/**", "/login").permitAll()
            .requestMatchers("/admin/**").hasRole("ADMIN")        // checks authority ROLE_ADMIN
            .requestMatchers(HttpMethod.GET, "/reports/**").hasAuthority("report:read")
            .anyRequest().authenticated())                        // or denyAll() to fail closed
        .formLogin(form -> form.loginPage("/login").permitAll()) // omit loginPage for the generated page
        .httpBasic(Customizer.withDefaults())
        .logout(Customizer.withDefaults());                       // POST /logout (+ CSRF token)
    return http.build();
}

@Bean
PasswordEncoder passwordEncoder() {                               // {bcrypt} by default
    return PasswordEncoderFactories.createDelegatingPasswordEncoder();
}

@Bean
static RoleHierarchy roleHierarchy() {                            // admins also pass hasRole("USER")
    return RoleHierarchyImpl.fromHierarchy("ROLE_ADMIN > ROLE_USER");
}
```

Plus a `UserDetailsService` bean (in-memory in the snippet; JPA/JDBC-backed in real apps). Defining it (or any `SecurityFilterChain`) makes Boot's generated user and default chain back off (§10.1).

JWT resource server (stateless API, first of two chains)

```
@Bean
@Order(1)                                                         // specific chain first
SecurityFilterChain api(HttpSecurity http) throws Exception {
    http.securityMatcher("/api/**")                               // which requests this chain owns
        .authorizeHttpRequests(auth -> auth
            .requestMatchers(HttpMethod.GET, "/api/**").hasAuthority("SCOPE_orders.read")
            .requestMatchers(HttpMethod.POST, "/api/**").hasAuthority("SCOPE_orders.write")
            .anyRequest().authenticated())
        .oauth2ResourceServer(rs -> rs.jwt(Customizer.withDefaults()))   // issuer-uri from properties
        .sessionManagement(s -> s.sessionCreationPolicy(SessionCreationPolicy.STATELESS))
        .csrf(csrf -> csrf.disable());                            // no cookies on this chain
    return http.build();
}
```

```
spring.security.oauth2.resourceserver.jwt.issuer-uri=https://idp.example.com/realms/shop
spring.security.oauth2.resourceserver.jwt.audiences=orders-api
# optional: roles from a claim, without writing a converter
spring.security.oauth2.resourceserver.jwt.authorities-claim-name=roles
spring.security.oauth2.resourceserver.jwt.authority-prefix=ROLE_
```

Note the last two lines replace the default `scope` → `SCOPE_` mapping; to keep both, write a converter that combines two `JwtGrantedAuthoritiesConverter`s (§11.5). The browser chain (no `securityMatcher`) gets `@Order(2)` — catch-all chains go last (§10.3).

Method security

```
@Configuration
@EnableMethodSecurity                                          // @PreAuthorize & co. (proxy-based)
class MethodSecurityConfig { }

public class OrderService {
    @PreAuthorize("hasRole('ADMIN') or #owner == authentication.name")
    public List<Order> ordersOf(String owner) { ... }

    @PostAuthorize("returnObject.owner() == authentication.name")
    public Order find(long id) { ... }

    @PreAuthorize("hasAuthority('order:refund')")
    public String refund(long id) { ... }
}
```

Proxy rules apply: self-invocation and `private` methods are **not** protected (§10.7).

CORS and CSRF for a cookie-based single-page app

```
@Bean
SecurityFilterChain spa(HttpSecurity http) throws Exception {
    http.authorizeHttpRequests(auth -> auth.anyRequest().authenticated())
        .cors(Customizer.withDefaults())         // uses the CorsConfigurationSource bean below
        .csrf(csrf -> csrf.spa())                 // Security 7: XSRF-TOKEN cookie + X-XSRF-TOKEN header
        .formLogin(Customizer.withDefaults());
    return http.build();
}

@Bean
CorsConfigurationSource corsConfigurationSource() {
    CorsConfiguration api = new CorsConfiguration();
    api.setAllowedOrigins(List.of("https://app.example.com"));   // exact origins, never "*" with credentials
    api.setAllowedMethods(List.of("GET", "POST", "PUT", "DELETE"));
    api.setAllowedHeaders(List.of("Content-Type", "X-XSRF-TOKEN"));
    api.setAllowCredentials(true);
    api.setMaxAge(3600L);
    UrlBasedCorsConfigurationSource source = new UrlBasedCorsConfigurationSource();
    source.registerCorsConfiguration("/api/**", api);
    return source;
}
```

The rules in one table

Fact	Detail	§
`hasRole("ADMIN")`	checks the authority `ROLE_ADMIN` ✔; `hasRole("ROLE_ADMIN")` throws `IllegalArgumentException` ✔	§10.6
`hasAuthority("x")`	checks the literal string (`report:read`, `SCOPE_orders.read`)	§10.6
JWT scopes	`scope`/`scp` claim → `SCOPE_x` authorities	§11.5
401 vs 403	anonymous + denied → 401 (or login redirect); authenticated + denied → 403	§10.6
JWT failures	bad/expired token → 401 `invalid_token`; missing scope → 403 `insufficient_scope`; 60 s clock skew	§11.3
CSRF	on for unsafe methods; disable only on chains without cookies/cached Basic	§10.8
CORS	`http.cors(...)` so preflights are answered before authentication	§10.9
sessions	`IF_REQUIRED` (default), `ALWAYS`, `NEVER`, `STATELESS`; session id changes at login	§10.10
passwords	`DelegatingPasswordEncoder`: `{bcrypt}` by default ✔, reads `{noop}`, `{argon2...}`; `upgradeEncoding` → re-hash on login ✔	§10.5
MFA (S7)	every login adds a `FACTOR_*` authority (`FACTOR_PASSWORD`, `FACTOR_OTT`, `FACTOR_WEBAUTHN`)	§10.4
removed in S7	`.and()`, `authorizeRequests`, `AntPathRequestMatcher`/`MvcRequestMatcher`, OAuth2 password grant client support	§17.15
security context	`SecurityContextHolder` is a `ThreadLocal`: use `DelegatingSecurityContext*` executors for `@Async`	§10.2

19.9 Testing annotations and slices

Whole-application tests

Annotation / type	Module / test starter	Purpose	§
`@SpringBootTest`	spring-boot-test (`spring-boot-starter-test`)	full context; `webEnvironment = MOCK` (default), `RANDOM_PORT`, `DEFINED_PORT`, `NONE`	§12.6
`@AutoConfigureMockMvc`	`-webmvc-test`	`MockMvc` + `MockMvcTester` in a `@SpringBootTest` (no longer automatic in Boot 4)	§12.6
`@AutoConfigureRestTestClient`	spring-boot-resttestclient (in `-webmvc-test`)	inject `RestTestClient` (F7), bound to MockMvc or the running server	§12.9
`@AutoConfigureWebTestClient`	`-webflux-test`	`WebTestClient` in a `@SpringBootTest`	§13.12
`@AutoConfigureTestRestTemplate`	spring-boot-resttestclient	legacy `TestRestTemplate`	§12.9
`@LocalServerPort`	spring-boot-test	the random port	§1.12
`@TestConfiguration`	spring-boot-test	extra beans; nested static = added, top-level = `@Import` it	§12.6
`@ActiveProfiles`	spring-test	activate profiles (part of the cache key)	§12.15
`@TestPropertySource` / `properties = ...`	spring-test / spring-boot-test	test-only properties	§12.15
`@DynamicPropertySource`	spring-test	runtime values via `Supplier`s; highest precedence	§12.12
`@DirtiesContext`	spring-test	close and evict the cached context (avoid)	§12.11
`@Sql`	spring-test	run SQL scripts before/after a test	§12.16
`@Transactional` (on a test)	spring-test	rolled back after each test — can hide transaction bugs	§12.7, §12.17

Slices

Slice	Test starter	Loads	Rolled back?	Extras	§
`@WebMvcTest(Controller.class)`	`spring-boot-starter-webmvc-test`	MVC, Jackson, validation, security filters, your controllers/advice/filters	no	`MockMvcTester`, `MockMvc`	§12.7
`@WebFluxTest`	`spring-boot-starter-webflux-test`	WebFlux, codecs, controllers	no	`WebTestClient`	§13.10
`@DataJpaTest`	`spring-boot-starter-data-jpa-test`	JPA, repositories, embedded DB, Flyway/Liquibase	**yes**	`TestEntityManager`, SQL shown	§12.7
`@JdbcTest`	`spring-boot-starter-jdbc-test`	`DataSource`, `JdbcClient`, `JdbcTemplate`	**yes**	—	§7.6
`@DataJdbcTest`	`spring-boot-starter-data-jdbc-test`	Spring Data JDBC + Flyway	**yes**	—	§9.2
`@DataR2dbcTest`	`spring-boot-starter-data-r2dbc-test`	R2DBC repositories	**no**	—	§13.14
`@JsonTest`	`spring-boot-starter-jackson-test`	Jackson `JsonMapper` with your customisations	no	`JacksonTester`	§12.7
`@RestClientTest(Client.class)`	`spring-boot-starter-restclient-test`	`RestClient.Builder`, Jackson, the named client	no	`MockRestServiceServer`	§6.8
`@DataMongoTest` / `@DataRedisTest`	`...-data-mongodb-test` / `...-data-redis-test`	the store's repositories	no	—	§9.7
`@AutoConfigureTestDatabase(replace = ...)`	`spring-boot-starter-jdbc-test`	`NON_TEST` (default since 3.4) replaces your DB unless it is already a test DB; `NONE` keeps it	—	—	§12.7

Mocks, containers, security helpers

Annotation / helper	Module	Purpose	§
`@MockitoBean`	spring-test (F6.2)	Mockito mock **in the context** (replaced removed `@MockBean`); reset after each test	§12.10
`@MockitoSpyBean`	spring-test	spy around the real bean (replaced `@SpyBean`)	§12.10
`@TestBean`	spring-test	override a bean with a static factory method's result	§12.10
`@Mock` + `@ExtendWith(MockitoExtension.class)`	Mockito	plain unit test, no Spring	§12.4
`@Testcontainers` / `@Container`	Testcontainers 2	start Docker containers for the test	§12.14
`@ServiceConnection`	spring-boot-testcontainers	take connection details from the container (no URL properties)	§12.14
`@EmbeddedKafka`	spring-kafka-test (`-kafka-test`)	in-JVM KRaft broker; sets `spring.embedded.kafka.brokers`	§14.13
`@AutoConfigureMetrics` / `@AutoConfigureTracing`	`-micrometer-metrics-test` / `-opentelemetry-test`	keep metric export / span export on in tests	§15.5, §15.7
`@WithMockUser` / `@WithUserDetails` / `@WithAnonymousUser`	spring-security-test	identity for a test method	§10.12
`user()`, `httpBasic()`, `csrf()`, `formLogin()`, `logout()`	spring-security-test	request post-processors / builders	§10.12
`jwt()`, `opaqueToken()`, `oauth2Login()`, `oidcLogin()`	spring-security-test	token identities — pass your converter to `jwt()`	§11.6
`ApplicationContextRunner`	spring-boot-test	tiny context per test, for auto-configuration and conditions	§3.13
`StepVerifier` (+ `withVirtualTime`)	reactor-test	verify `Mono`/`Flux` signals	§13.5
Awaitility `await().atMost(...)`	awaitility (in `spring-boot-starter-test`)	wait for asynchronous results, never `Thread.sleep`	§14.2

Test clients and the context cache

Client	Stack	Needs a server?	§
`MockMvcTester` (F6.2, AssertJ style)	MVC	no	§12.8
`RestTestClient` (F7)	MVC (or any server with `bindToServer`)	optional	§12.9
`WebTestClient`	WebFlux (also MockMvc)	optional	§13.12
`TestRestTemplate`	any	yes	§12.9

Context cache: key = the merged configuration (classes, profiles, properties, `@MockitoBean`s, `@DynamicPropertySource`, `@AutoConfigure...`, `webEnvironment`); **max 32** contexts, LRU (least recently used — the context unused for longest is evicted first) (`spring.test.context.cache.maxSize`); Framework 7 **pauses** inactive contexts (`spring.test.context.cache.pause`). Keep one shared base class (§12.11).

19.10 Reactor operators quick table

`Mono` = 0..1 item, `Flux` = 0..N items; both end with complete **or** error, and **nothing runs until someone subscribes** (in WebFlux, the framework does) (§13.3).

Goal	Operator(s)	Remember	§
create from values	`Mono.just(v)`, `Flux.just(...)`, `Flux.fromIterable(list)`, `Flux.range(1, n)`	`just(expr)` evaluates `expr` **immediately**	§13.3
create lazily	`Mono.fromCallable(...)`, `Mono.defer(() -> ...)`, `Mono.fromFuture(...)`	runs per subscription — retries repeat it	§13.3
empty / error	`Mono.empty()`, `Mono.error(e)`		§13.3
timers	`Flux.interval(d)`, `Mono.delay(d)`, `delayElements(d)`	run on `Schedulers.parallel()`	§13.3
bridge callbacks	`Flux.create(sink -> ...)`, `Sinks.many()`, `Sinks.one()`	choose an overflow strategy	§13.9
transform (sync)	`map`	must not block or return a publisher	§13.4
async call per item	`flatMap(fn[, concurrency])`	concurrent, **interleaved** output; default concurrency 256	§13.4
async, one at a time, in order	`concatMap`	sequential (sum of times)	§13.4
async, concurrent, source order	`flatMapSequential`	buffers early results	§13.4
filter	`filter`, `take(n)`, `skip(n)`, `distinct`		§13.4
combine by position	`Mono.zip(a, b, c)`, `zipWith`	"call 3 services in parallel" (like `Promise.all`)	§13.4
combine in sequence / as they come	`concat`, `concatWith` / `merge`, `mergeWith`		§13.4
fallback when empty	`switchIfEmpty(...)`, `defaultIfEmpty(v)`	the usual 404 path	§13.4
aggregate	`collectList`, `collectMap`, `reduce`, `count`	return a `Mono`	§13.4
group	`buffer(n)`, `window(n)`, `groupBy`		§13.4
then	`then()`, `then(mono)`	"do A, then B", ignoring A's values	§13.4
side effects	`doOnNext`, `doOnError`, `doOnSubscribe`, `doFinally`, `log()`	not for business logic	§13.4
errors: fallback	`onErrorReturn(v)`, `onErrorResume(e -> pub)`, `onErrorMap(e -> ex)`	an error **ends** the sequence	§13.6
errors: retry	`retry(n)`, `retryWhen(Retry.backoff(3, Duration.ofSeconds(1)))`	re-subscribes; attempts at 0/1/3/7 s with `jitter(0)`	§13.6
timeout	`timeout(d[, fallback])`	before `retryWhen` = per attempt	§13.6, §13.13
backpressure	`limitRate(n)` (re-requests at 75 %), `onBackpressureDrop/Latest/Buffer(n)/Error`	unbounded buffers risk `OutOfMemoryError`	§13.7
threads	`publishOn(s)` (everything **after** it), `subscribeOn(s)` (the **source**; position irrelevant)		§13.8
schedulers	`Schedulers.parallel()`, `boundedElastic()` (wrap blocking calls), `single()`, `immediate()`	never block Netty event loops	§13.8
hot / shared	`share()`, `publish().autoConnect()`, `cache()`, `Sinks.many().multicast()`	late subscribers miss earlier items	§13.9
context	`contextWrite(c -> c.put(k, v))`, `Mono.deferContextual(ctx -> ...)`	the reactive `ThreadLocal`	§13.9
leave the reactive world	`block()`, `blockFirst()`, `blockLast()`	tests and `main` only — throws on event-loop threads	§13.3
test	`StepVerifier.create(pub).expectNext(...).verifyComplete()`, `withVirtualTime(() -> ...)` + `thenAwait(d)`	build the publisher **inside** the supplier	§13.5

19.11 Actuator endpoints

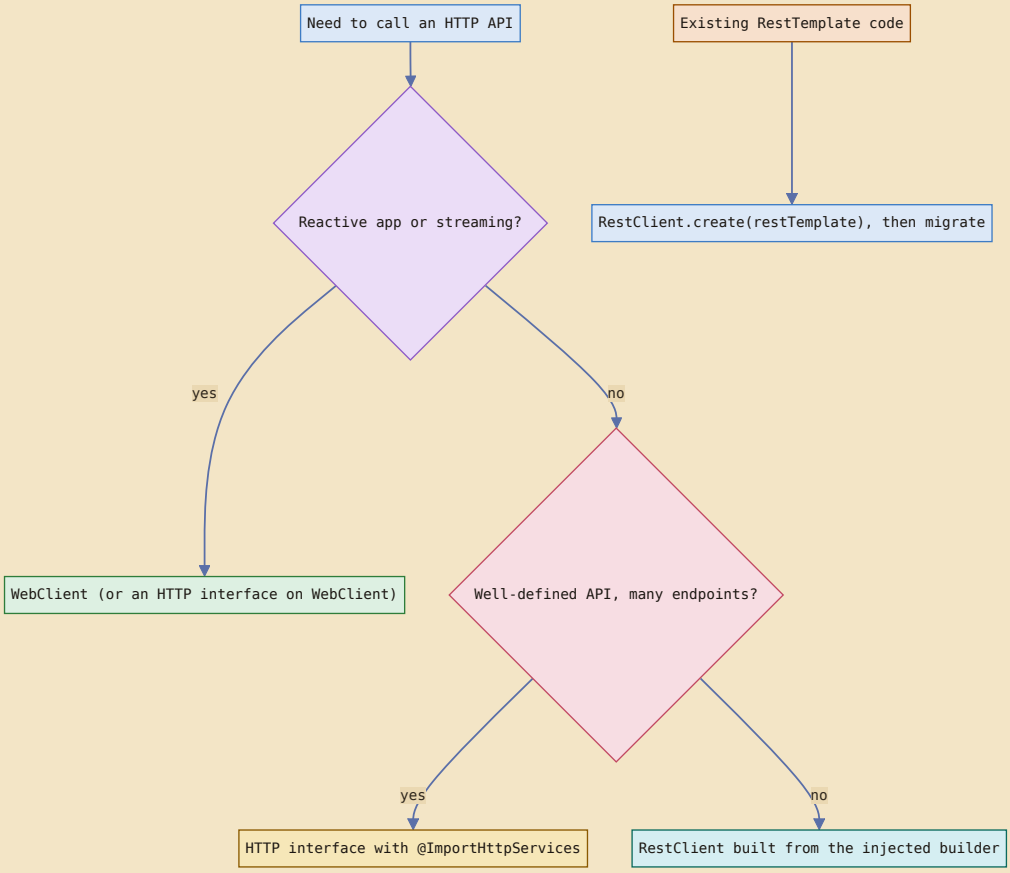
Add `spring-boot-starter-actuator`. Two switches decide availability: **access** (`none` / `read-only` / `unrestricted`) and **exposure** (HTTP / JMX — Java Management Extensions, the JVM's built-in management channel). By default only `health` is exposed over HTTP (§15.2).

Endpoint (`/actuator/<id>`)	Shows / does	Notes	§
`health`	status `UP` / `DOWN` / `OUT_OF_SERVICE` / `UNKNOWN`; groups `liveness`, `readiness` (on by default in Boot 4)	`DOWN` / `OUT_OF_SERVICE` → HTTP 503; details `never` by default	§15.3
`info`	build, git, java, os, process, `info.*`	contributors switched on by properties	§15.4
`metrics`, `metrics/{name}`	meters as JSON, drill down with `?tag=k:v`	for humans	§15.5
`prometheus`	all meters in Prometheus text format	needs `micrometer-registry-prometheus`	§15.5
`loggers`	view and change log levels at runtime	`POST` with `{"configuredLevel":"DEBUG"}`	§15.2
`env`, `configprops`	property sources, bound `@ConfigurationProperties`	values masked (`show-values=never`)	§3.11, §15.2
`conditions`	the auto-configuration report	"why was bean X (not) created?"	§3.11
`beans`, `mappings`	all beans / all request mappings	debugging	§15.2
`threaddump`, `heapdump`	JVM threads / heap dump file	heap dumps can contain secrets	§15.2
`scheduledtasks`, `caches`, `flyway`, `liquibase`, `sessions`, `quartz`	feature-specific views (`caches` can also clear)	appear when the feature is present	§15.2
`httpexchanges`	recent requests/responses	needs an `HttpExchangeRepository` bean	§15.2
`startup`	startup steps and durations	needs `BufferingApplicationStartup`	§15.2
`sbom`, `auditevents`, `integrationgraph`, `logfile`	software bill of materials, audit events, Spring Integration graph, the log file	when configured	§15.2
`shutdown`	stops the app	access `none` by default	§15.2
`refresh`, `busrefresh` (Spring Cloud)	re-bind config / broadcast a refresh	Spring Cloud Context / Bus	§16.3

Secure them with `EndpointRequest.to("health", "info")` / `EndpointRequest.toAnyEndpoint()` (Boot 4 package `org.springframework.boot.security.autoconfigure.actuate.web.servlet`), ideally on a separate `management.server.port` (§15.2).

19.12 Which tool for which job

HTTP client: `RestClient` vs `WebClient` vs HTTP interface



	`RestClient`	`WebClient`	**HTTP interface (`@HttpExchange`)**	`RestTemplate`
Style	fluent, blocking	fluent, non-blocking (`Mono` / `Flux`)	declarative Java interface → proxy	template methods
Since / status	F6.1, recommended for blocking code	F5.0, recommended for reactive	F6.0; Boot 4 groups via `@ImportHttpServices`	deprecation announced (F7.1 `@Deprecated` , removal in 8.0)
Starter (Boot 4)	`spring-boot-starter-restclient`	`spring-boot-starter-webclient`	`-restclient` (or `-webclient` with `clientType = WEB_CLIENT`)	`-restclient`
Pick when	hand-written calls, custom per-call logic	WebFlux apps, streaming, heavy async composition	typed clients to other services, least boilerplate	never for new code
Watch out	inject `RestClient.Builder` (Boot converters, timeouts, tracing); `exchange()` skips status handlers	don't `block()` on event loops	wrap for `Optional` /domain exceptions	—
§	§6.2	§13.13	§6.6	§6.9

Web stack: Spring MVC + virtual threads vs WebFlux

	MVC + `spring.threads.virtual.enabled=true`	WebFlux
Code style	plain imperative, blocking libraries OK	`Mono` / `Flux` pipelines, non-blocking everything
Measured (1,000 clients, 100 ms I/O)	**7,942 req/s**, p50 108 ms, 33 platform threads	**8,058 req/s**, p50 110 ms
Same test, MVC on 200 platform threads	1,909 req/s, p50 517 ms	WebFlux + one `Thread.sleep`: **97 req/s**
Data access	JDBC, JPA	R2DBC, reactive Mongo/Redis
Streaming, backpressure end to end (SSE, WebSockets)	possible (`SseEmitter`), each stream holds a request	natural (`Flux`)
Debugging, `ThreadLocal` libraries	easy	needs Reactor `Context`, context propagation
Choose for	typical CRUD/REST services (the default)	streaming, gateways, all-non-blocking pipelines
§	§13.15, §15.10	§13.15

Data access: JPA vs `JdbcClient` vs Spring Data JDBC

	Spring Data JPA (Hibernate)	`JdbcClient` / `JdbcTemplate`	Spring Data JDBC
Model	rich object graph, relationships both ways	your own SQL, rows → records	DDD aggregates: root + owned children
Magic	persistence context, lazy loading, dirty checking, first-level cache	none	none: `save` writes, loading reads the whole aggregate
SQL you can predict	not always (N+1, flush timing)	yes	yes (children are deleted + re-inserted on save)
Schema	`ddl-auto` possible (use Flyway)	you write it	you write it (Flyway)
Typical pitfalls	N+1, `LazyInitializationException`, OSIV	SQL injection by concatenation, `.single()` on 0 rows	assigned ids without `@Version` → silent `UPDATE` of 0 rows
Choose for	domain models with relationships, team knows JPA	reports, bulk work, full SQL control	clear aggregates, predictable SQL
§	§8.1	§7.3	§9.2

Also: R2DBC for non-blocking SQL (§13.14), MongoDB for flexible documents (§9.5), Redis for counters/sessions/shared cache (§9.6); the overview table is §9.9.

Cache: Caffeine vs Redis

	Caffeine (local)	Redis (shared)
Read cost	nanoseconds, no serialization	network hop + (de)serialization
Consistency across instances	each instance has its own copy (stale until TTL)	one copy for all
Survives restarts	no	yes (with Redis persistence)
Stores	object **references** — cache immutable values	serialized copies (choose a JSON serializer)
Stampede protection	`@Cacheable(sync = true)` per key per JVM	same annotation, still per JVM; distributed lock for cluster-wide
TTL / size	`expireAfterWrite`, `maximumSize` (`spring.cache.caffeine.spec`)	`spring.cache.redis.time-to-live` / `RedisCacheConfiguration`
Choose for	small, hot, rarely changing reference data	data shared between instances, sessions, expensive results
§	§9.8	§9.8

Messaging: Kafka vs RabbitMQ

	Kafka	RabbitMQ
Model	partitioned, replicated **log**; consumers track offsets	**queue** broker: exchanges → bindings → queues; acked messages are deleted
Replay / many independent readers	yes: each consumer group has its own offset	no: each reader needs its own queue
Ordering	per partition (use the aggregate id as key)	per queue, weakened by competing consumers
Routing	topic + partition only	rich: direct, topic, fanout, headers
Per-message TTL, priority, delay	no	yes (delay via plugin)
Spring	`KafkaTemplate`, `@KafkaListener`, `DefaultErrorHandler` → `<topic>-dlt`, `@RetryableTopic`	`RabbitTemplate`, `@RabbitListener`, DLX, publisher confirms
Test without Docker	`@EmbeddedKafka`	no embedded broker (Testcontainers)
Choose for	event streaming, event sourcing, high throughput	task distribution, complex routing, request/reply
§	§14.9	§14.18

Resilience: Resilience4j vs Framework 7 `@Retryable`

Need	Framework 7 core	Resilience4j	Spring Retry
Retry with back-off + jitter	`@Retryable` (`maxRetries` = retries **after** the first call ✔), `RetryTemplate`	`@Retry` (`max-attempts` counts the first call)	yes (no longer managed by Boot 4)
Bulkhead	`@ConcurrencyLimit` (`BLOCK` / `REJECT`)	`@Bulkhead` (semaphore / thread pool)	no
Circuit breaker	**no**	`@CircuitBreaker` (CLOSED → OPEN → HALF_OPEN)	basic
Rate limiter / time limiter	no	`@RateLimiter`, `@TimeLimiter`	no
Fallback methods	no (catch at the caller)	`fallbackMethod`	`@Recover`
Metrics, health	`MethodRetryEvent`, `RetryListener`	Micrometer meters, health indicators	listeners
Switch	`@EnableResilientMethods`	`resilience4j-spring-boot4` + properties	`@EnableRetry`
Choose for	simple retries and concurrency caps, no extra dependency	circuit breaking, rate limits, fallbacks	legacy code only
§	§4.7	§16.8	§4.7

Rules for any retry: only transient failures (short-lived ones that may succeed a moment later) and idempotent operations (safe to repeat), exponential back-off (longer waits after each failure) with jitter (a random extra delay so clients don't retry in lockstep), a total time budget, retry in **one** layer (§6.7).

More decisions in one line each

Question	Answer	§
`@Value` or `@ConfigurationProperties`?	one value / SpEL → `@Value`; a group of settings → a `@ConfigurationProperties` record	§3.6
`@Component` or `@Bean`?	your class → stereotype; third-party class or creation logic → `@Bean`	§2.3
Unit test, slice or `@SpringBootTest`?	logic → unit; Spring-produced behaviour of one layer → slice; wiring → a few `@SpringBootTest`s	§12.2
`Page` or `Slice` (or `Window`)?	page numbers → `Page` (extra COUNT); "load more" → `Slice`; big tables → keyset `Window`	§8.11
`@EventListener`, `@TransactionalEventListener` or outbox?	in-process now → `@EventListener`; only after commit → `@TransactionalEventListener`; must not be lost → outbox	§2.12, §7.15, §14.17
JWT or opaque token?	local fast validation → JWT; instant revocation / hidden claims → opaque + introspection	§11.7
Session cookie or tokens?	browser app / BFF → session (+ CSRF); service-to-service, mobile → bearer tokens	§10.10, §11.12
Blocking or non-blocking retries (Kafka)?	order matters → `DefaultErrorHandler` (blocking); independent records → `@RetryableTopic`	§14.15
Flyway or Liquibase?	SQL-first, one database → Flyway; rollbacks, preconditions, many databases → Liquibase	§9.4
Eureka or Kubernetes Services?	on Kubernetes, Services + DNS usually suffice	§16.4
OpenFeign or HTTP interfaces?	new code → HTTP interfaces (OpenFeign is in maintenance mode)	§16.6
JVM + AOT cache or native image?	long-running, throughput-heavy → JVM (AOT cache halves startup); CLIs, scale-to-zero → native	§15.11

19.13 Boot 3 → 4 quick migration checklist

Work top to bottom; each step names the symptom you see if you skip it. Details: §17.12, §17.13, §17.14, §17.15.

Before you start

- [ ] Upgrade to **Boot 3.5.x** first and remove every deprecation warning (4.0 deletes what 3.x deprecated). From 2.7, go 2.7 → 3.5 (`javax` → `jakarta`) first.
- [ ] Java 17+ (25 recommended); Kotlin 2.2+, GraalVM 25+, Gradle 8.14+ if you use them.
- [ ] Run OpenRewrite's (an automated code-refactoring tool) `org.openrewrite.java.spring.boot4.UpgradeSpringBoot_4_0` recipe and review the diff.
- [ ] Add `spring-boot-properties-migrator` (runtime, temporarily) to report renamed properties; remove it afterwards.
- [ ] Stuck? `spring-boot-starter-classic` / `spring-boot-starter-test-classic` restore the old all-in-one layout while you fix imports.

Dependencies

- [ ] `spring-boot-starter-web` → `spring-boot-starter-webmvc`; `-aop` → `-aspectj`; `-oauth2-*` → `-security-oauth2-*`; `-web-services` → `-webservices`.
- [ ] Add `spring-boot-starter-restclient` (client code) and `spring-boot-starter-webclient` (`WebClient.Builder`, else "No qualifying bean of type WebClient$Builder").
- [ ] Add a **test starter** per technology (`-webmvc-test`, `-data-jpa-test`, ...) — else `@WebMvcTest` does not resolve.
- [ ] Add `spring-boot-starter-flyway` / `-liquibase` — else migrations silently stop running.
- [ ] Undertow → Tomcat or Jetty. Spring Retry → declare its version or move to `@Retryable` (F7).
- [ ] Separately versioned projects: Spring Cloud **2025.1.x** (2025.1.2+ for Boot 4.1), Spring AI **2.0**, springdoc **3.x**, Testcontainers **2.x** artifact names.

Code

- [ ] Jackson 3: `com.fasterxml.jackson.databind` → `tools.jackson.databind` (annotations unchanged); `ObjectMapper` bean → `JsonMapper`; `Jackson2ObjectMapperBuilderCustomizer` → `JsonMapperBuilderCustomizer`; `@JsonComponent` → `@JacksonComponent`.
- [ ] Auto-configuration class imports moved to per-module packages (check `exclude = ...`, `@ImportAutoConfiguration`, `@AutoConfiguration(after = ...)`).
- [ ] `@EntityScan` → `org.springframework.boot.persistence.autoconfigure`; `hibernate-jpamodelgen` → `hibernate-processor`.

- [ ] Null-safety: `org.springframework.lang.Nullable` → `org.jspecify.annotations.Nullable`; `javax.annotation`/`javax.inject` (the old Java EE package names, now `jakarta.*`) are gone.
- [ ] Security 7: rewrite `.and()` chains as one lambda per feature, `authorizeRequests` → `authorizeHttpRequests`, Ant/MVC matchers → `PathPatternRequestMatcher` (plain `requestMatchers("/x/**")`).
- [ ] `RestTemplate` → `RestClient` (start with `RestClient.create(restTemplate)`).

Tests

- [ ] `@MockBean` / `@SpyBean` → `@MockitoBean` / `@MockitoSpyBean` (removed in Boot 4; not allowed inside `@Configuration` classes).
- [ ] `@SpringBootTest` + `@AutoConfigureMockMvc` / `@AutoConfigureRestTestClient` / `@AutoConfigureWebTestClient` / `@AutoConfigureTestRestTemplate` — no longer automatic.
- [ ] Slice annotation imports moved (e.g. `org.springframework.boot.webmvc.test.autoconfigure.WebMvcTest`); JUnit 4 runners → JUnit 6 (Jupiter).

Properties (most common renames)

Boot 3	Boot 4
`spring.jackson.parser.*` / `generator.*`	`spring.jackson.json.read.*` / `json.write.*`
`spring.data.mongodb.uri` / `.host` / ...	`spring.mongodb.uri` / `.host` / ...
`spring.http.client.*`	`spring.http.clients.*`
`server.error.*` (e.g. `include-stacktrace`, `path`)	`spring.web.error.*`
`spring.dao.exceptiontranslation.enabled`	`spring.persistence.exceptiontranslation.enabled`
`management.tracing.enabled`	`management.tracing.export.enabled`
`management.otlp.tracing.endpoint`	`management.opentelemetry.tracing.export.otlp.endpoint`
`management.health.mongo.enabled`	`management.health.mongodb.enabled`
`spring.session.redis.*`	`spring.session.data.redis.*`
`spring.kafka.retry.topic.backoff.random`	`spring.kafka.retry.topic.backoff.jitter`
`spring.rabbitmq.listener.simple.retry.max-attempts` (counted the first call)	`...retry.max-retries` (does not)

Behaviour changes to re-test

- [ ] Health probes (`/actuator/health/liveness`, `/readiness`) are **on by default** — check what your platform calls.
- [ ] DevTools LiveReload is off (and deprecated in 4.1).
- [ ] `server.forward-headers-strategy` no longer applies to WAR deployments (apps packaged as a `.war` file and deployed into an external servlet container).
- [ ] Jackson 3 defaults: ISO dates, unknown properties ignored, properties sorted alphabetically after creator properties.

19.14 Interview index

Chapter 18 holds 168 questions with short answers, five runnable proofs (small tests that demonstrate an answer by running it), a project walkthrough and ten debugging scenarios. This index maps each topic to its Chapter 18 section and to the chapter that teaches it.

Topic	Chapter 18 section	Questions	Teaching chapter
IoC, beans, scopes, lifecycle, events	§18.2	Q1–Q20	Chapter 2
Boot, starters, configuration, auto-configuration	§18.3	Q21–Q35	Chapter 1, Chapter 3
AOP, proxies, Framework 7 resilience	§18.4	Q36–Q44	Chapter 4
MVC, REST, validation, errors, versioning, clients	§18.5	Q45–Q61	Chapter 5, Chapter 6
JDBC, transactions, JPA, Spring Data	§18.6	Q62–Q91	Chapter 7, Chapter 8, Chapter 9
Caching	§18.7	Q92–Q97	Chapter 9
Security, JWT, OAuth2	§18.8	Q98–Q114	Chapter 10, Chapter 11
Testing	§18.9	Q115–Q125	Chapter 12
Reactive and WebFlux	§18.10	Q126–Q132	Chapter 13
Messaging, async, scheduling	§18.11	Q133–Q140	Chapter 14
Production and observability	§18.12	Q141–Q152	Chapter 15
Microservices and Spring Cloud	§18.13	Q153–Q160	Chapter 16
Spring AI and Boot 4 / Framework 7 changes	§18.14	Q161–Q168	Chapter 17
Proofs of five tricky answers	§18.15	prototype-in-singleton, `@Qualifier` vs `@Primary`, cycles, checked exceptions, self-invocation	this chapter's §19.15 adds more
"Explain your Spring project" template	§18.16	—	—
Ten debugging scenarios	§18.17	bean not found, cycles, no rollback, lazy loading, N+1, 401/403, CORS, "annotation does nothing", auto-configuration failures, slow startup	—

The twenty questions that come up most

#	Question	Short answer	Study
1	IoC vs DI?	framework creates objects / objects receive dependencies	§2.1
2	`BeanFactory` vs `ApplicationContext`?	lazy minimal container / eager, events, environment, post-processors	§2.2
3	Why constructor injection?	`final`, complete objects, testable with `new`, cycles fail fast	§2.4
4	Two beans of one type?	`@Qualifier` > `@Primary` > `@Fallback` > parameter name	§2.5
5	Singleton thread-safe? prototype in singleton?	shared → keep stateless / created once, use `ObjectProvider`	§2.6
6	Bean lifecycle?	§19.7 order	§2.7
7	How does auto-configuration work?	`.imports` files, conditions, applied after your beans, `@ConditionalOnMissingBean`	§3.10
8	Property precedence?	command line > system props > env vars > profile files > `application.*`	§3.3
9	JDK vs CGLIB proxy; self-invocation?	interfaces vs subclass (Boot default CGLIB) / `this.x()` skips the proxy	§4.4, §4.5
10	Request flow in MVC?	filters → `DispatcherServlet` → mapping → interceptors → arguments → controller → converter → resolvers	§5.2
11	Global error handling?	`@RestControllerAdvice` + `ProblemDetail` (RFC 9457)	§5.7
12	How does `@Transactional` work?	proxy + manager + thread-bound connection	§7.7
13	Propagation types; checked exception?	§19.6 / commits by default	§7.9, §7.10
14	N+1 and `LazyInitializationException`?	JOIN FETCH / `@EntityGraph` / batch size; fetch per use case, DTOs	§8.7, §8.6
15	Security filter chain; 401 vs 403?	`DelegatingFilterProxy` → `FilterChainProxy` → first matching chain	§10.2, §10.6
16	JWT validation in a resource server?	signature via JWKS, `exp` / `nbf` (60 s skew), `iss`, `aud`, `scope` → `SCOPE_`	§11.3
17	Slices vs `@SpringBootTest`; context caching?	partial vs full context / cache keyed by configuration, max 32	§12.7, §12.11
18	WebFlux or virtual threads?	MVC + virtual threads for CRUD; WebFlux for streaming	§13.15
19	Reliable DB write + event?	transactional outbox + idempotent consumers	§14.17
20	What's new in Boot 4 / Framework 7?	modular starters, Jackson 3, JSpecify, API versioning, HTTP service clients, `@Retryable`, `RestTestClient`	§17.13, §17.14

19.15 Proofs: the facts on these sheets, run

What it is

`code/cheatsheet-checks/` is a small Maven (Java's standard build tool) project (Boot 4.1.1, Java 25, no web server, no Docker) whose tests assert the facts marked ✔ in this chapter. If a future Boot version changes a default, a test fails, and the sheet must be updated. Run it from the repository root with `bash tools_run_spring.sh spring '19-*'`.

Test class	Proves	Sheet
`ScopeAndLifecycleTests`	singleton shared, prototype fresh, prototype-in-singleton stuck, init/destroy order, no destroy callback for prototypes	§19.7
`BeanSelectionTests`	`@Qualifier` beats `@Primary`, `@Primary` beats the parameter name, `List` by `@Order`, `Map` by bean name	§19.3, §19.7
`TransactionTests`	`@Transactional` defaults, all seven propagation types, rollback rules, `UnexpectedRollbackException`	§19.6
`PropertyBindingTests`	env-var mapping (canonical and legacy), list index, `Duration` / `DataSize` units, record + `@DefaultValue`	§19.5
`PropertyPrecedenceTests`	defaults < `application.properties` < env var < `-D` < `--arg`; `@Value` defaults and SpEL	§19.5
`BootDefaultsTests`	HikariCP pool 10 / timeout 30 s, unique embedded DB name, `@Async` executor 8 threads `task-` (virtual threads → `SimpleAsyncTaskExecutor`), scheduler 1 thread, shutdown 30 s	§19.5
`CoreUtilitiesTests`	default cache keys, six-field cron, `@Retryable` defaults and 1 + `maxRetries` attempts	§19.3, §19.12
`JsonAndSecurityDefaultsTests`	Jackson 3 defaults, `hasRole` prefix rules, `DelegatingPasswordEncoder`	§19.5, §19.8

The transaction checks use plain Spring on an in-memory H2 database (a small Java SQL database that lives only in the test's memory), the same technique as §18.15:

```
@Transactional
void catchInnerRequired() {
    inner.insert("outer");
    try { inner.requiredThenFail(); } catch (IllegalStateException swallowed) { }
}   // commit attempt -> UnexpectedRollbackException, nothing is saved
```

The environment-variable checks bind through the same `Binder` that `@ConfigurationProperties` uses, over a fake `SystemEnvironmentPropertySource`:

```
Binder canonical = envBinder(Map.of("SHOP_CHECKOUTTIMEOUT", "2m", "SHOP_PAYMENT_RETRIES", "4", "SHOP_CURRENCIES_0", "USD"));
Duration timeout = canonical.bind("shop.checkout-timeout", Duration.class).get();   // PT2M
```

Output

`mvn -q -B -ntp verify` (20 tests, all passing; JUnit does not fix the order of test classes, so blocks may appear in another order):

```
spring.task.scheduling.pool.size default = 1, spring.lifecycle.timeout-per-shutdown-phase default = PT30S
applicationTaskExecutor = ThreadPoolTaskExecutor, core = 8, prefix = task-, queue = unbounded
with spring.threads.virtual.enabled=true -> SimpleAsyncTaskExecutor
DataSource = HikariDataSource, maximumPoolSize = 10, connectionTimeout = 30000 ms
no spring.datasource.url + H2 on the classpath -> jdbc:h2:mem:<random name>  (generate-unique-name=true)
'0 30 2 * * MON-FRI' after 2026-09-28T10:00 -> 2026-09-29T02:30
five-field Unix cron '0 2 * * *' -> IllegalArgumentException
SimpleKeyGenerator: () -> SimpleKey [], ("BOOK-1") -> BOOK-1, ("BOOK-1","EUR") -> SimpleKey [BOOK-1, EUR]
@Retryable defaults: maxRetries=3, delay=1000 ms, multiplier=1.0
maxRetries = 2 -> [attempt 1, attempt 2, attempt 3], then the last exception is rethrown
SHOP_CHECKOUTTIMEOUT=2m   -> shop.checkout-timeout = PT2M
SHOP_PAYMENT_RETRIES=4    -> shop.payment.retries  = 4
SHOP_CURRENCIES_0=USD     -> shop.currencies[0]    = USD
SHOP_CHECKOUT_TIMEOUT=3m  -> shop.checkout-timeout = PT3M (legacy '-' -> '_' form, also accepted)
Duration '45s' -> PT45S, '500' -> 500 ms, 'PT1H' -> PT1H
DataSize '10MB' -> 10485760 bytes, '1024' -> 1024 bytes
record bound with @DefaultValue: ShopProperties[name=Corner Shop, checkoutTimeout=PT30S, maxUpload=5242880B]
init order    : [constructor, @PostConstruct, afterPropertiesSet, initMethod]
destroy order : [@PreDestroy, destroy(), destroyMethod]  (no Proto @PreDestroy)
singleton: getBean twice -> same instance?        true
prototype: getBean twice -> new instance?         true
prototype injected into a singleton -> same one?  true
@Value ${shop.missing:fallback} -> fallback, #{${shop.retries:3} + 1} with retries=2 -> 3, key set only in defaultProperties -> eu
(default)
defaultProperties + application.properties -> from application.properties
+ env SHOP_NAME                            -> from env var
+ -Dshop.name                              -> from -D system property
+ --shop.name                              -> from --command-line
by type (@Primary)        -> sms
@Qualifier("email")        -> email
parameter named 'push'    -> sms  (@Primary wins over the name)
List<Notifier> by @Order  -> [email, sms, push]
Map keys (bean names)     -> [email, push, sms]
encode -> {bcrypt}$2a$10$...  matches {noop}s3cret? true  upgradeEncoding({noop}) = true
hasRole("ADMIN"): authority ROLE_ADMIN -> true, authority ADMIN -> false
hasRole("ROLE_ADMIN") -> IllegalArgumentException
JsonMapper (tools.jackson.databind.json) writes {"item":"book","on":"2026-09-28"}
unknown property ignored -> Delivery[item=pen, on=2026-10-01]
broken JSON -> JacksonException (unchecked)
committed despite an exception: [checked, default rules, caught inside the method]
inner REQUIRED failed, outer caught it -> UnexpectedRollbackException, rows = []
@Transactional defaults: propagation=REQUIRED isolation=DEFAULT readOnly=false timeout=-1
REQUIRED      : inner runs in the outer transaction? true (Outer.requiredJoins)
REQUIRES_NEW  : outer rolled back, rows left = [audit (REQUIRES_NEW)]
NESTED        : inner failed + caught, rows left = [outer] (savepoint)
MANDATORY     : no transaction -> IllegalTransactionStateException
NEVER         : inside a transaction -> IllegalTransactionStateException
SUPPORTS      : active alone? false, active inside a transaction? true
NOT_SUPPORTED : active inside a transaction? false (outer suspended)
```

What the output teaches

- A **legacy** environment-variable spelling (`SHOP_CHECKOUT_TIMEOUT`, dash turned into an underscore) also binds to `shop.checkout-timeout`. The documented form without the dash separator (`SHOP_CHECKOUTTIMEOUT`) is the one to use and to search for; know that both work when you hunt for "who set this value?".
- `@Primary` beats a matching parameter name — the name is only the **last** tie-breaker.
- HikariCP reports its pool size only after the pool has started; the test takes one connection first.

- The `@Retryable` bean is a CGLIB proxy, so the test reads its attempt list through a method: a field read on a proxy returns the proxy's own, never-initialised field (§4.5).

19.16 Glossary A–Z

Short definitions of the terms used in this book. Annotations are listed without `@` in the sort order. Follow the link for the full explanation.

A

Term	Meaning	See
Access (Actuator)	per-endpoint permission: `none`, `read-only`, `unrestricted` (Boot 3.4+); separate from exposure	§15.2
Access token	short-lived token a client sends to an API to prove it may call it	§11.1
`acks` (Kafka)	how many replicas must confirm a write; `all` (`-1`) is the default since Kafka 3.0	§14.16
Actuator	Boot's management endpoints: health, metrics, info, loggers, env...	§15.2
Advice (AOP)	the code an aspect runs at a join point: before, after returning, after throwing, after, around	§4.1
Advisor (AOP)	Spring's internal pair of one advice and one pointcut	§4.1
Advisor (Spring AI)	interceptor around every `ChatClient` call (memory, RAG, tool calling, logging), ordered by `getOrder()`	§17.5
AES-GCM	standard symmetric encryption (one shared key) with built-in tamper detection; used for encrypting secrets and data	§10.5
Aggregate	DDD cluster of objects loaded and saved as one unit through its root; the unit of Spring Data JDBC	§9.2
`AggregateReference`	Spring Data JDBC reference to another aggregate by id only	§9.2
Annotation (Java)	metadata written as `@Name` on a class, method or field; Spring reads it to add behaviour	§1.1
Anthropic / OpenAI / Ollama	AI model providers Spring AI has starters for; Ollama runs open models on your own machine	§17.2
AOP	Aspect-Oriented Programming: write a cross-cutting concern once, apply it where a pointcut matches	§4.1
AOT cache (JDK)	JDK 24+ file of classes loaded and linked in a training run; `-XX:AOTCacheOutput` / `-XX:AOTCache`	§15.11
AOT processing (Spring)	build-time generation of bean-definition code, used by native images and optionally on the JVM	§15.11
API	Application Programming Interface: the operations one piece of software offers to another (Java methods or HTTP endpoints)	§1.1
API gateway	single front door that routes requests and applies edge concerns (Spring Cloud Gateway)	§16.7
API versioning	Framework 7 feature: read a version from the request and match `version = "1.1+"` mappings	§5.9
`ApiVersionInserter`	client-side strategy that puts the API version into requests	§6.7
`ApplicationContext`	Spring's full container: eager singletons, post-processors, events, `Environment`, i18n	§2.2
`ApplicationContextRunner`	test helper that builds a tiny context per test, ideal for auto-configuration	§3.13
`ApplicationEventPublisher`	bean used to publish application events	§2.12
`ApplicationReadyEvent`	Boot event after startup and runners have finished	§2.7
`ApplicationRunner` / `CommandLineRunner`	beans run once at startup with the program arguments	§1.8
Argon2 / bcrypt / PBKDF2 / scrypt	slow, salted password-hashing algorithms; bcrypt is Spring Security's default	§10.5
Aspect	class bundling pointcuts and advice for one concern (`@Aspect` + `@Component`)	§4.2
AspectJ	full AOP language with compile/load-time weaving; Spring borrows its annotations and pointcut syntax	§4.8
AssertJ	fluent assertion library: `assertThat(x).isEqualTo(y)`; included in the Boot test starter	§12.8
Asynchronous	the caller continues without waiting; the result arrives later (callback, future, message)	§14.1
At-least-once	delivery guarantee: never lost, possibly duplicated — consumers must be idempotent	§14.8
Audience (`aud`)	JWT claim naming who the token is meant for; an API should reject tokens for another audience	§11.1
Auditing (Spring Data)	automatically filling created/modified date and user columns	§8.13
`Authentication`	object holding the principal, credentials and authorities of the current caller	§10.1
Authentication	proving who the caller is	§10.1
`AuthenticationEntryPoint`	starts authentication for an anonymous, denied request: 401 challenge or login redirect	§10.2
`AuthenticationManager`	entry point of authentication; usually a `ProviderManager` asking `AuthenticationProvider`s	§10.4
Authority (`GrantedAuthority`)	one permission string held by the caller (`report:read`, `ROLE_ADMIN`, `SCOPE_x`)	§10.1
Authorization	deciding whether a known caller may perform an action	§10.1
Authorization code flow	OAuth2 flow for apps with a user: code in the browser, token exchange on the back channel (with PKCE)	§11.9

Term	Meaning	See
Authorization server	issues tokens (Keycloak, Okta, Entra ID, Spring Authorization Server)	§11.2
`AuthorizationFilter`	last filter of a security chain; applies `authorizeHttpRequests` rules	§10.2
Auto-configuration	conditional `@Configuration` classes that Boot applies after yours to fill gaps	§3.10
`AutoConfiguration.imports`	file `META-INF/spring/...AutoConfiguration.imports` listing a jar's auto-configurations	§3.10
Availability state	Boot's `LivenessState` and `ReadinessState`, exposed as health groups	§15.3
Awaitility	test library that polls a condition until true or timeout	§14.2
`Aware` callbacks	interfaces (`BeanNameAware`, `ApplicationContextAware`) through which a bean receives container objects	§2.7

B

Term	Meaning	See
B3 headers	Zipkin's older trace-propagation headers, an alternative to W3C `traceparent`	§15.7
Backpressure	the consumer limits the producer via `request(n)`	§13.7
Baggage	extra key/value context that travels with a trace	§15.7
Bean	object created, wired and managed by the Spring container	§2.1
Bean definition	the recipe for a bean: class or factory method, scope, dependencies, callbacks	§2.2
Bean Validation	Jakarta standard for constraint annotations (`@NotBlank`, `@Email`); Hibernate Validator implements it	§5.6
`BeanFactory`	minimal container: bean definitions + lazy creation	§2.2
`BeanFactoryPostProcessor`	changes bean **definitions** before any bean is created (e.g. `${}` resolution)	§2.8
`BeanPostProcessor`	sees every bean **instance** before/after init and may wrap it in a proxy	§2.8
`BeanRegistrar`	Framework 7 contract for registering beans programmatically (loops, conditions)	§2.14
Bearer token	token that grants access to whoever holds it, sent as `Authorization: Bearer ...`	§11.1
BFF (Backend for Frontend)	server-side OAuth2 client that keeps tokens away from the browser and gives it a session cookie	§11.12
BlockHound	Java agent that detects blocking calls on non-blocking threads in tests	§13.8
Blocking call	a call that holds its thread while it waits (JDBC, `Thread.sleep`, a blocking HTTP client)	§13.1
BOM	"bill of materials" POM pinning versions of many artifacts (`spring-boot-dependencies`)	§1.6
`boundedElastic`	Reactor scheduler for wrapping unavoidable blocking calls	§13.8
BPP / BFPP	short for `BeanPostProcessor` / `BeanFactoryPostProcessor`	§2.8
Brave / Zipkin / Jaeger / Tempo	tracing libraries and back ends: Brave records spans, Zipkin, Jaeger and Tempo store and display traces	§15.7
Broker (message broker)	server that receives messages from producers and delivers them to consumers (Kafka, RabbitMQ)	§14.8
Brute-force protection	slowing or locking out repeated failed logins so passwords can't simply be guessed	§11.15
Bucket4j	Java rate-limiting library based on the token-bucket algorithm	§11.15
Build tool	program that downloads dependencies, compiles, tests and packages the code (Maven, Gradle)	§1.6
Bulkhead	cap on concurrent calls to one dependency (`@ConcurrencyLimit`, Resilience4j `@Bulkhead`)	§4.7, §16.8
Buildpacks	Cloud Native Buildpacks: build an OCI image from a jar without a Dockerfile (`spring-boot:build-image`)	§15.11
Bytecode	the compiled instructions in `.class` files that the JVM executes	§4.4

C

Term	Meaning	See
Cache-aside	look in the cache; on a miss load, store and return — what `@Cacheable` does	§9.8
Cache stampede	many concurrent misses for one hot key all compute it; `sync = true` mitigates per JVM	§9.8
`CacheManager`	provider-neutral cache interface behind the cache annotations	§9.8
Caffeine	high-performance local Java cache (W-TinyLFU eviction)	§9.8
Cardinality	number of distinct tag-value combinations of a metric; keep it small	§15.5
Cascade	passing a JPA operation (persist, merge, remove...) from parent to children	§8.5
Cascading failure	one slow or failed service makes its callers fail too, spreading through the system	§16.8
CDC (Change Data Capture)	streaming every insert/update/delete from the database's change log (e.g. Debezium); one way to relay an outbox	§14.17
CDS	Class Data Sharing: JVM archive of parsed classes for faster startup	§15.11
CGLIB proxy	runtime subclass used as a proxy; Boot's default; cannot handle `final`	§4.4
`ChatClient`	Spring AI fluent API over a `ChatModel`, with advisors, templates, tools	§17.3
`ChatMemory`	Spring AI store of past messages per conversation id (window of 20 by default)	§17.6
`ChatModel`	Spring AI provider-level interface: `Prompt` in, `ChatResponse` out	§17.3
Checked exception	exception the compiler forces you to declare or catch (not a `RuntimeException`); commits a transaction by default	§7.10
Chunking (RAG)	splitting documents into small pieces before embedding them, so retrieval returns focused passages	§17.9
Circuit breaker	stops calling a failing dependency (CLOSED → OPEN → HALF_OPEN)	§16.8
Circular dependency	dependency cycle between beans; constructor cycles always fail, Boot rejects field cycles	§2.9
Claim	statement in a JWT payload (`sub`, `iss`, `aud`, `exp`, `scope`)	§11.1
Class loader	JVM component that finds and loads `.class` files; DevTools uses two of them for fast restart	§1.10
Classpath	the list of jars and folders the JVM loads classes from; auto-configuration reacts to what is on it	§1.2
CLI	command-line interface: a program you run from a terminal	§1.9
Client credentials	OAuth2 grant for service-to-service calls without a user	§11.10
Cold publisher	replays the whole pipeline for every subscriber	§13.9
Commit / rollback	finish a transaction by making its changes permanent / by undoing all of them	§7.7
Component scanning	finding `@Component` classes in a package tree	§2.3
`ConcurrencyLimit`	Framework 7 annotation limiting concurrent calls of a method (`BLOCK` or `REJECT`)	§4.7
Condition evaluation report	Boot's list of which auto-configurations matched and why (`--debug`, `/actuator/conditions`)	§3.11
`Conditional`	register a bean only if a `Condition` matches; base of all `@ConditionalOn...`	§2.11
Config data	Boot's files and imports: `application.*`, profile files, `spring.config.import`	§3.3
Config Server	Spring Cloud service serving configuration from Git or other backends	§16.3
`configtree:`	import where each file in a directory is a property (Kubernetes secrets)	§3.8
`ConfigurationProperties`	binds a prefix of properties into a typed, validated object	§3.6
Connection pool	set of open database connections reused across requests because opening one is slow (HikariCP)	§7.2
Constructor injection	dependencies passed as constructor parameters; the recommended injection style	§2.4
Consul	HashiCorp service registry and key/value store, an alternative to Eureka	§16.4
Consumer group	Kafka consumers sharing a `group.id`; each partition goes to one member, each group gets every record	§14.9
Consumer lag	how many messages a consumer group is behind the newest ones in a partition	§14.12
Container (Docker)	isolated process packaged with everything it needs to run; started from a container image	§12.14
Container image	packaged file system plus start command from which containers are run (OCI format)	§15.11
Content negotiation	choosing the representation from `Accept` / `produces` (406 if impossible)	§5.8
Context caching (tests)	the TestContext Framework reuses started contexts with the same configuration	§12.11

Term	Meaning	See
Context pausing	Framework 7: inactive cached test contexts have their lifecycle beans stopped	§12.11
Context propagation	copying `ThreadLocal` values (trace, MDC) across threads or into the Reactor `Context`	§15.6, §13.9
Contract testing	consumer and provider tested against a shared contract (Pact, Spring Cloud Contract)	§16.10
Cookie	small value a site stores in the browser, sent back automatically on every request to that site	§10.10
CORS	browser rules letting a page from one origin read responses from another (preflight `OPTIONS`)	§10.9
Counter / Timer / DistributionSummary	Micrometer meters: a count that only rises / durations / sizes	§15.5
CPU-bound vs I/O-bound	work limited by computation vs work that mostly waits for network or disk	§13.8
CQRS	Command Query Responsibility Segregation: separate models for writes and for reads	§16.1
Cron expression	Spring's six-field schedule, starting with seconds (`0 0 2 * * *`)	§14.7
Cross-cutting concern	requirement spread over many classes (logging, transactions, security)	§4.1
CRUD	Create, Read, Update, Delete: the four basic data operations	§8.8
CSRF	attack that makes a logged-in browser send a forged state-changing request; blocked by tokens	§10.8
curl / Postman	command-line and graphical tools for sending HTTP requests by hand; they ignore CORS	§1.5
Customizer	bean that adjusts an auto-configured object instead of replacing it	§3.12

D

Term	Meaning	See
DAO	Data Access Object: a class that hides the SQL for one kind of data	§7.1
`DataAccessException`	Spring's unchecked, database-neutral exception hierarchy	§7.5
`DataSize`	Boot type for sizes (`10MB`); plain numbers are bytes	§3.6
`DataSource`	factory of JDBC connections; in Boot a HikariCP pool	§7.2
DDD	Domain-Driven Design: modelling software around business concepts (entities, aggregates, bounded contexts)	§9.2
DDL / `ddl-auto`	Data Definition Language (`CREATE TABLE`, `ALTER`); `ddl-auto` lets Hibernate create or validate the schema	§8.16
Dead-letter exchange (DLX)	RabbitMQ exchange that receives rejected or expired messages from a queue	§14.18
Dead-letter topic (DLT)	where records go after the last failed attempt; default name `<topic>-dlt` (spring-kafka 3.3+)	§14.14
`DeadLetterPublishingRecoverer`	spring-kafka recoverer that publishes failed records to the DLT	§14.14
Deadlock	two transactions each wait for a lock the other holds; the database aborts one of them	§7.11
`DefaultErrorHandler`	spring-kafka container error handler: blocking retries, then a recoverer	§14.14
`DelegatingFilterProxy`	servlet filter that hands requests to the `springSecurityFilterChain` bean	§10.2
`DelegatingPasswordEncoder`	encoder storing `{id}hash`; verifies old formats, upgrades on login	§10.5
Dependency	a library your project uses, declared in the build file; or an object a bean needs	§1.6
Dependency Injection (DI)	an object receives its dependencies instead of creating them	§2.1
Deprecated	still works but is marked for removal in a later version; move off it	§1.4
Derived query	Spring Data method whose name is the query (`findByTitleContaining`)	§8.9
Deserialization / serialization	turning bytes or JSON into objects / objects into bytes or JSON	§5.5
Detached entity	JPA entity no longer in a persistence context; changes are not tracked	§8.4
DevTools	development module: fast restart, dev defaults; off for `java -jar`	§1.10
Dirty checking	Hibernate compares managed entities with their snapshots at flush and writes `UPDATE`s	§8.4
`DirtiesContext`	test annotation that closes and evicts the cached context	§12.11
`DiscoveryClient`	Spring Cloud abstraction returning the instances of a service id	§16.4
`DispatcherServlet`	Spring MVC's front controller	§5.2
`DispatcherHandler`	WebFlux's front controller	§13.10
DNS	Domain Name System: turns host names into network addresses; Kubernetes uses it for service discovery	§16.4
Docker	tool that builds and runs containers; Testcontainers needs it	§12.14
DTO	Data Transfer Object: a simple class or record shaped for an API instead of the database	§5.4
Dual-write problem	a database write and a broker publish cannot be atomic; solved by the outbox	§14.17
`Duration` binding	`45s`, `5m`, `PT1H`; plain numbers are milliseconds	§3.6
`DynamicPropertySource`	test method registering properties with lazy suppliers (random ports, containers)	§12.12

E

Term	Meaning	See
EAGER / LAZY	load an association with its owner / on first access; make everything LAZY	§8.6
EJB	Enterprise JavaBeans: the older Java EE component model Spring was created as an alternative to	§1.4
Embedded database	database that runs inside the application's JVM (H2, HSQLDB); handy for tests and demos	§7.2
Embedded server	Tomcat/Jetty/Netty running inside the application jar	§1.2
`EmbeddedKafka`	in-JVM Kafka broker for tests (spring-kafka-test)	§14.13
Embedding	vector of numbers representing text so similar texts are close	§17.9
`EnableAutoConfiguration`	imports Boot's auto-configuration candidates	§3.10
`EnableResilientMethods`	Framework 7 switch for `@Retryable` and `@ConcurrencyLimit`	§4.7
Entity	JPA class whose instances are table rows (`@Entity`, `@Id`, no-arg constructor, not `final`)	§8.3
Entity graph	declares associations to fetch with a query (`@EntityGraph`, left join)	§8.7
`EntityManager`	JPA API over the persistence context (Hibernate `Session`)	§8.4
`Environment`	Spring's view of all property sources and active profiles	§3.1
ETag	response header holding a version fingerprint so clients can ask "has it changed?" (304 Not Modified)	§8.14
Eureka	Netflix service registry supported by Spring Cloud (AP design)	§16.4
Event loop	few threads that run pipeline steps when I/O completes, never waiting	§13.1
Event-driven	services communicate by publishing and reacting to events instead of calling each other directly	§14.8
`EventListener`	method receiving application events, synchronously by default	§2.12
Eventual consistency	copies or services agree after a short delay instead of instantly	§14.17
Exactly-once	each message's effect happens once; in practice at-least-once delivery plus idempotent processing	§14.16
Exception translation	turning `SQLException`s into `DataAccessException`s	§7.5
`ExceptionTranslationFilter`	turns security exceptions into 401/redirect (anonymous) or 403 (authenticated)	§10.2
Exchange / binding (RabbitMQ)	an exchange receives messages and routes them to queues according to bindings	§14.18
Executor / `TaskExecutor`	owner of worker threads and a task queue; runs `@Async` work	§14.4
Exponential back-off	waiting longer after each failed attempt (1 s, 2 s, 4 s...) before retrying	§4.7
Exposure (Actuator)	whether an endpoint is reachable over HTTP or JMX (`health` only, by default)	§15.2
Externalized configuration	settings outside the build artifact (files, env vars, args)	§3.1

F

Term	Meaning	See
`FACTOR_*` authority	Security 7 authority recording a completed login factor (`FACTOR_PASSWORD`)	§10.4
`Fallback`	Framework 6.2: bean used only if no other candidate qualifies	§2.5
Fallback (resilience)	default answer returned when a protected call fails or a circuit is open	§16.8
Fat (executable) jar	Boot jar with `BOOT-INF/lib` dependencies, started by `JarLauncher`	§1.9
Filter (servlet)	code around every request before the `DispatcherServlet`; can wrap request/response	§5.11
`FilterChainProxy`	the `springSecurityFilterChain` bean that picks the first matching `SecurityFilterChain`	§10.2
`fixedDelay` / `fixedRate`	`@Scheduled` timing: end-to-start / start-to-start	§14.7
`flatMap` / `concatMap`	async per item: concurrent and interleaved / sequential and ordered	§13.4
Flush	Hibernate sends queued SQL: before commit, before affected queries, on `flush()`	§8.4
`Flux`	Reactor publisher of 0..N items	§13.3
Flyway	SQL-file schema migration tool (`V1__x.sql`, checksums)	§8.16
Foreign key	column that points to the primary key of a row in another table	§8.5
Form login	username/password via an HTML form; session cookie afterwards	§10.4
Full vs lite configuration	`@Configuration` CGLIB subclass (inter-bean calls return singletons) vs plain calls	§2.3
Functional endpoints	routes as `RouterFunction` beans and handler functions instead of annotations	§13.11

G – H

Term	Meaning	See
GA / milestone / SNAPSHOT	release stages: general availability (production-ready) / preview builds / unreleased development builds	§1.4
Garbage collection (GC)	the JVM automatically freeing memory of objects no longer used; pauses can add latency	§13.15
Gateway (Spring Cloud)	API gateway: routes = predicates + filters; Web MVC or WebFlux flavour	§16.7
Gauge	Micrometer meter sampling a current value	§15.5
GPU	graphics processor; runs AI models much faster than a CPU	§10.5
Graceful shutdown	stop accepting requests, finish in-flight ones, then close (default since Boot 3.4)	§15.9
GraalVM native image	whole app compiled ahead of time to a native executable (closed world)	§15.11
Gradle	build tool alternative to Maven, with Groovy or Kotlin build scripts	§1.5
Grant type (OAuth2)	way a client obtains a token: authorization code, client credentials, refresh token, token exchange	§11.2
H2	small Java SQL database that can run in memory; used for demos and quick tests	§7.2
HAL / HATEOAS	REST style where responses include links to related actions; HAL is a JSON format for those links	§5.12
`HandlerInterceptor`	MVC hook around handler methods (`preHandle`, `postHandle`, `afterCompletion`)	§5.11
`HandlerMapping`	finds the handler method for a request	§5.2
Hash (password)	one-way fingerprint of a value; passwords are stored as slow, salted hashes, never encrypted	§10.5
Health group	named subset of health indicators at `/actuator/health/<group>`	§15.3
`HealthIndicator`	bean contributing one check to `/actuator/health`	§15.3
Heap	JVM memory area where objects live; sized with `-Xmx` or `-XX:MaxRAMPercentage`	§15.11
Hibernate	the JPA implementation (ORM) Boot uses; maps entities to tables and generates SQL	§8.1
HikariCP	Boot's default JDBC connection pool (default 10 connections)	§7.2
HMAC / RSA / EC (JWT signing)	signing families: HMAC (HS256) shares one secret; RSA (RS256) and EC (ES256) sign privately and verify with a public key	§11.1
Hot publisher	produces whether or not anyone listens; late subscribers miss items	§13.9
HOTP / TOTP	one-time password algorithms based on a counter / on the current time	§11.18
HSQLDB / Derby	other embedded Java databases Boot can auto-configure, like H2	§7.2
HTTP	HyperText Transfer Protocol: request/response protocol of the web (method, path, headers, body, status)	§5.1
HTTP Basic	credentials in an `Authorization: Basic` header on every request	§10.4
HTTP methods	GET reads, POST creates, PUT replaces, PATCH partly updates, DELETE removes	§5.1
HTTP service interface	Java interface with `@GetExchange`... implemented by a generated proxy	§6.6
HTTP status 401 vs 403	401 = not authenticated (who are you?), 403 = authenticated but not allowed	§10.6
`HttpMessageConverter`	converts bodies to and from Java objects (Jackson for JSON)	§5.2
`HttpOnly` / `Secure` / `SameSite`	cookie flags: hidden from JavaScript / sent only over HTTPS / not sent on cross-site requests	§11.12
HTTPS / TLS	HTTP encrypted with TLS (Transport Layer Security), which also proves the server's identity	§11.18

I – J

Term	Meaning	See
ID token	OIDC JWT that tells the client who logged in; not for calling APIs	§11.2
IDE	Integrated Development Environment such as IntelliJ IDEA, Eclipse or VS Code	§1.5
Idempotency key	unique id a client sends with a request so a retried request is processed only once	§5.1
Idempotent	repeating the operation gives the same result (GET, PUT, DELETE; not POST)	§5.1
Idempotent consumer	de-duplicates messages by id so at-least-once delivery is safe	§14.17
Idempotent producer	Kafka producer whose retries cannot duplicate records (on by default)	§14.16
Identity provider (IdP)	system that owns user accounts and issues tokens	§11.2
`IDENTITY` / `SEQUENCE`	JPA id strategies: auto-increment (no insert batching) / database sequence	§8.3
`ImportHttpServices`	Framework 7 registration of HTTP interface groups; Boot configures them by properties	§6.6
Integration test	test of several real parts working together (often a whole context plus a real database)	§12.2
Interceptor	code that runs around a call (method, handler or HTTP request) and can change or stop it	§5.11
Introspection	asking the authorization server whether an opaque token is active (RFC 7662)	§11.7
Inversion of Control (IoC)	the framework, not your code, creates objects and calls you	§2.1
Isolation level	how much a transaction sees of concurrent ones (READ COMMITTED...)	§7.11
Issuer (`iss`)	JWT claim naming the authorization server that created the token	§11.1
J2EE / Java EE	old names of the enterprise Java standards, now Jakarta EE	§1.4
Jackson 3	Boot 4's JSON library, packages `tools.jackson.*`	§5.5
Jakarta EE	successor of Java EE; `jakarta.*` packages (EE 11 in Boot 4)	§1.4
Jar / WAR	Java archive of compiled classes / web archive deployed into an external servlet container	§1.9
`JarLauncher`	`Main-Class` of a Boot fat jar that loads nested jars and calls your `Start-Class`	§1.9
Java record	compact immutable data class (`record Order(long id, String status)`); ideal for DTOs and properties	§1.11
JDBC	Java Database Connectivity: Java's standard low-level API for SQL databases	§7.1
`JdbcClient`	Framework 6.1 fluent API for SQL with automatic record mapping	§7.3
`JdbcTemplate`	classic SQL template; still used for batch updates	§7.3
JDK	Java Development Kit: compiler, tools and the Java runtime	§1.4
JDK dynamic proxy	`java.lang.reflect.Proxy` implementing the bean's interfaces only	§4.4
JEP	JDK Enhancement Proposal: the numbered design document for a new Java feature	§13.15
JIT compiler	Just-In-Time compiler: turns frequently run bytecode into machine code while the app runs	§15.11
Jitter	random variation added to retry delays so clients don't retry in lockstep	§4.7
JMS	Jakarta Messaging: the standard Java API for message brokers such as ActiveMQ Artemis	§14.18
JMX	Java Management Extensions: built-in JVM channel for monitoring and management tools	§15.2
Join point	a point where advice can run; in Spring AOP always a bean method execution	§4.1
`JOIN FETCH`	JPQL that loads an association in the same query (N+1 fix)	§8.7
JPA	Jakarta Persistence specification; Hibernate implements it	§8.1
JPQL	query language over entities and fields, not tables	§8.10
JSON	JavaScript Object Notation: the text format of most REST APIs (`{"id": 1}`)	§5.5
`JsonMapper`	Jackson 3's immutable JSON mapper, built with `JsonMapper.builder()`	§5.5
JSpecify	standard null-safety annotations (`@NullMarked`, `@Nullable`) used by Framework 7	§2.13
JUnit	the standard Java test framework (JUnit 6 / Jupiter in Boot 4)	§12.3
JVM	Java Virtual Machine: the runtime that loads and executes bytecode	§1.2
JWK Set (JWKS)	JSON document of an issuer's public keys, selected by `kid`	§11.1
JWS / JWE	a signed JWT (contents readable, tamper-proof) / an encrypted JWT (contents hidden)	§11.1
JWT	signed token `header.payload.signature`; claims readable by anyone	§11.1
`JwtAuthenticationConverter`	turns JWT claims into authorities (`scope` → `SCOPE_x`)	§11.5

Term	Meaning	See
`JwtDecoder`	parses and validates a JWT (signature, `exp`, `iss`, custom validators)	§11.3

K – L

Term	Meaning	See
Kafka	distributed, replicated commit log for event streaming	§14.9
`KafkaListener`	spring-kafka consumer method	§14.12
`KafkaTemplate`	spring-kafka producer API; `send` returns a `CompletableFuture`	§14.11
Key (Kafka)	decides the partition; same key → same partition → ordered	§14.9
Keycloak	open-source identity provider and authorization server	§11.2
KMS	Key Management Service: cloud service that stores encryption keys and performs crypto without revealing them	§11.13
KRaft	Kafka's built-in consensus replacing ZooKeeper (Kafka 4.0+)	§14.9
Kubernetes (K8s)	platform that runs, restarts, scales and connects containers across machines	§15.3
Lambda (Java)	short inline function, `x -> x * 2`; the Security DSL is configured with lambdas	§10.3
Latency	how long one request takes from start to finish (often reported as p50/p99)	§15.5
Layered jar	Boot jar split into layers (dependencies, application code) so container images rebuild only what changed	§15.11
Lazy initialization	create beans on first use (`@Lazy`, `spring.main.lazy-initialization`)	§2.10
`LazyInitializationException`	touching a lazy association after its persistence context closed	§8.6
LDAP	directory protocol used by companies to store users and groups; Spring Security can authenticate against it	§9.1
Lifecycle callbacks	init/destroy hooks: `@PostConstruct`, `afterPropertiesSet`, `@PreDestroy`...	§2.7
Liquibase	changelog-based migration tool with rollbacks and preconditions	§9.4
Liveness / readiness / startup probes	Kubernetes health checks: restart me? / send me traffic? / am I still starting?	§15.3
Liveness probe	"restart me?" — must depend only on internal state	§15.3
LLM	large language model: the text-generating AI behind chat APIs	§17.1
Load balancer	component that spreads requests over several instances of a service	§16.5
Load balancing (client-side)	the caller picks an instance per request (Spring Cloud LoadBalancer, round robin)	§16.5
Load testing (Gatling, k6, JMeter)	tools that fire many concurrent requests to measure throughput and latency under load	§12.18
Logback / SLF4J	SLF4J is the logging API your code calls; Logback is Boot's default implementation behind it	§15.8
Lombok	library that generates getters, constructors, `equals` and similar code at compile time	§8.3
`Lookup`	method injection returning a fresh prototype per call	§2.6
Lost update	two read-modify-write cycles overwrite each other; prevent with SQL math, locks or `@Version`	§7.11
LRU	least recently used: an eviction policy that removes the entry unused for longest	§9.8

M

Term	Meaning	See
Managed entity	entity tracked by the persistence context; changes are flushed automatically	§8.4
`mappedBy`	marks the inverse side of a JPA relationship; only the owning side is written	§8.5
`MappedCollection`	Spring Data JDBC child collection mapping (`idColumn`, `keyColumn`)	§9.2
Maven	Java build tool: `pom.xml` declares dependencies and plugins; `mvn verify` builds and tests	§1.6
MCP	Model Context Protocol: standard way to expose and call AI tools across processes	§17.10
MD5 / SHA-256	fast hash functions; fine for checksums, unsafe alone for passwords because they are fast to guess	§10.5
MDC	Mapped Diagnostic Context: per-thread log fields (request id, trace id)	§14.4, §15.1
`merge` vs `persist`	copy a detached entity's state onto a managed copy / make a new entity managed	§8.4
`MeterRegistry`	Micrometer registry where meters are recorded and exported	§15.5
Method security	authorization on bean methods (`@PreAuthorize...`), proxy-based	§10.7
Method validation	constraints on method parameters; built into MVC since Framework 6.1	§5.6
Metrics / logs / traces	the three kinds of observability data: numbers over time / event records / request paths	§15.1
Micrometer	vendor-neutral metrics facade ("SLF4J for metrics")	§15.1
Micrometer Tracing	tracing facade over OpenTelemetry or Brave	§15.7
MFA	multi-factor authentication; Security 7 tracks factors as `FACTOR_*` authorities	§10.4
Microservice	small, independently deployed service owning one business capability and its data	§16.1
Mock / stub / spy	test doubles: a mock records and checks calls, a stub returns canned answers, a spy wraps a real object	§12.4
Mockito	the Java mocking library behind `@MockitoBean`	§12.10
`MockitoBean` / `MockitoSpyBean`	mock / spy registered in the test context (replaced `@MockBean` / `@SpyBean`)	§12.10
`MockMvc` / `MockMvcTester`	in-process MVC test client; `MockMvcTester` adds AssertJ assertions (F6.2)	§12.8
`MockRestServiceServer`	fake server bound to a `RestClient.Builder` in `@RestClientTest`	§6.8
Modular starters	Boot 4: one starter, one auto-configuration module and one test starter per technology	§3.14
MongoDB	document database storing JSON-like documents (BSON)	§9.5
`Mono`	Reactor publisher of 0..1 item	§13.3
Monolith / modular monolith	one deployable application; modular when its internal modules have strict boundaries (Spring Modulith)	§16.1
mTLS	mutual TLS: both client and server present certificates, so each proves its identity	§11.10
Multi-tenancy (JWT)	several issuers in one API; `JwtIssuerAuthenticationManagerResolver` with an allow-list	§11.8
MVCC	Multi-Version Concurrency Control: the database keeps row versions so readers don't block writers	§7.11

N – O

Term	Meaning	See
N+1 problem	1 query for parents + 1 per parent for a lazy association	§8.7
Native query	plain database-specific SQL in a Spring Data `@Query` instead of JPQL	§8.10
NDJSON	newline-delimited JSON: one JSON object per line, used for streaming responses	§13.10
`NESTED`	propagation using a savepoint inside the current transaction	§7.9
Netty	non-blocking network library; Reactor Netty is WebFlux's default server	§13.10
`NimbusJwtEncoder`	Spring Security's JWT signer; Security 7 adds `withKeyPair` / `withSecretKey` builders	§11.4
Non-blocking	a call that returns immediately and frees its thread; the result is delivered later	§13.1
Nonce	OIDC random value bound into the ID token against replay	§11.9
`NullMarked` / `Nullable`	JSpecify: non-null by default in a scope / this may be null	§2.13
OAuth 2.0	framework for delegated authorization with access tokens	§11.2
`ObjectProvider`	lazy/optional lookup of a bean (`getIfAvailable`, `getObject` for prototypes)	§2.5
Observation API	Micrometer API: instrument once, get a metric and a span	§15.6
`Observed`	annotation creating an observation around a method	§15.6
Offset (Kafka)	a record's position in its partition; committed per consumer group	§14.9
OIDC	OpenID Connect: identity layer on OAuth2 adding the ID token	§11.2
ONNX	portable file format for machine-learning models, e.g. local embedding models	§17.1
Opaque token	random token string checked by introspection; instantly revocable	§11.7
Open Session in View	persistence context kept open for the whole web request (`spring.jpa.open-in-view`)	§8.15
OpenAPI	machine-readable API description; generated by springdoc	§5.12
OpenFeign	Spring Cloud declarative client, in maintenance mode	§16.6
OpenRewrite	automated refactoring recipes used for Boot upgrades	§17.12
OpenTelemetry	open standard and SDK for traces/metrics; OTLP is its protocol	§15.7
Optimistic locking	`@Version` column checked in `UPDATE ... WHERE version = ?`	§8.14
ORM	Object-Relational Mapping: mapping classes to tables so you work with objects instead of SQL rows	§8.1
`orphanRemoval`	delete a child removed from its parent's collection	§8.5
OSIV	short for Open Session in View	§8.15
OTLP	OpenTelemetry Protocol: the standard wire format for exporting traces, metrics and logs	§15.7
Outbox (transactional)	write the event into an outbox table in the same transaction; a relay publishes it	§14.17
OWASP	Open Worldwide Application Security Project: publishes the Top 10 web security risks	§10.5
Owning side	the relationship side holding the foreign key; the only one JPA writes	§8.5

P

Term	Meaning	See
p99 / percentile	the value that 99% of measurements are below; p99 latency shows the slow tail	§15.5
Pact	polyglot consumer-driven contract-testing tool with a broker and `can-i-deploy` check	§16.10
`Page` / `Slice` / `Window`	paged result with total (extra COUNT) / "has next" only / keyset scrolling	§8.11
Partition	ordered, append-only log inside a Kafka topic; unit of parallelism	§14.9
Passkey / WebAuthn	password-less login with a key pair stored on the user's device; WebAuthn is the browser standard	§10.4
Password4j	hashing library behind the new password encoders of Spring Security 7	§10.5
`PathPattern`	Spring's URL pattern matcher (`{id}`, `**`, `{*path}`); no trailing-slash match	§5.3
`PathPatternRequestMatcher`	Security 7 request matcher using `PathPattern` syntax	§10.3
Persistence context	the `EntityManager`'s first-level cache and unit of work	§8.4
Pessimistic locking	`SELECT ... FOR UPDATE` via `@Lock(PESSIMISTIC_WRITE)`	§8.14
PKCE	proof key that binds an authorization code to the client that started the flow	§11.9
Platform thread	ordinary Java thread backed one-to-one by an operating-system thread	§13.1
`PlatformTransactionManager`	Spring interface that begins, commits and rolls back transactions	§7.7
Pod	smallest unit Kubernetes runs: one or more containers sharing a network address	§15.3
Pointcut	predicate selecting join points (`execution(...)`, `within(...)`, `@annotation(...)`)	§4.3
Poison pill	record that can never be deserialized; `ErrorHandlingDeserializer` sends it to the DLT	§14.10
POM	Project Object Model: Maven's `pom.xml` build file	§1.6
PostgreSQL	open-source relational database used for most examples in this book	§8.2
Precedence	which property source wins (command line > ... > defaults)	§3.3
Preflight	the browser's `OPTIONS` request asking CORS permission	§10.9
`Primary`	default bean when several candidates match	§2.5
Primary key	column (or columns) that uniquely identifies each row of a table	§8.3
Principal	the "who" of a request: user, service or device	§10.1
`ProblemDetail`	RFC 9457 error body: `type`, `title`, `status`, `detail`, `instance`	§5.7
Profile	named environment selecting beans (`@Profile`) and files (`application-dev.yaml`)	§3.7
Profile group	one profile name that activates several	§3.7
Projection	query result with only the needed fields (interface or record)	§8.12
Prometheus	monitoring system that scrapes metrics from `/actuator/prometheus` and stores them as time series	§15.5
Prompt template	text with `{placeholders}` filled at run time (Spring AI `PromptTemplate`)	§17.4
Propagation	what a transactional method does when a transaction already exists	§7.9
Property source	one named set of key/values inside the `Environment`	§3.1
Prototype scope	new bean instance per lookup; no destroy callbacks	§2.6
Proxy	stand-in object that runs extra behaviour and then calls the real bean	§4.4
`Proxyable`	Framework 7 per-bean choice of JDK or CGLIB proxy	§4.4
Publish/subscribe	messaging style where every subscriber gets its own copy of each message	§14.8
`publishOn` / `subscribeOn`	move downstream operators / the subscription to a scheduler	§13.8

Q – R

Term	Meaning	See
`Qualifier`	selects a specific bean at an injection point	§2.5
`QuestionAnswerAdvisor`	Spring AI advisor doing simple RAG over a `VectorStore`	§17.9
R2DBC	reactive (non-blocking) SQL API; not an ORM	§13.14
RabbitMQ	queue broker: exchanges route messages to queues via bindings	§14.18
RAG	retrieval-augmented generation: add retrieved documents to the prompt	§17.9
Rate limiter	cap on calls per time period (Resilience4j, gateway filter)	§16.8
Reactive programming	describing asynchronous data pipelines that are pushed through without blocking threads	§13.1
Reactive Streams	spec: `Publisher`, `Subscriber`, `Subscription`, `Processor` with demand	§13.2
Readiness probe	"send me traffic?" — may include essential dependencies	§15.3
Redis	in-memory key/value store used as a shared cache, session store or lock	§9.6
Reflection	Java code inspecting or calling classes, methods and fields at runtime	§2.4
Refresh (context)	`refresh()` creates all non-lazy singletons and starts the context	§2.2
Refresh token	long-lived token used only to get a new access token without logging in again	§11.1
`RefreshScope`	Spring Cloud scope re-creating beans after a configuration refresh	§16.3
Relaxed binding	several spellings map to one property (`checkout-timeout`, `checkoutTimeout`, `SHOP_CHECKOUTTIMEOUT`)	§3.4
Release train	set of versions released together (Spring Data 2026.0, Spring Cloud 2025.1)	§16.2
Replica / replication	a copy of data kept on another server; replication keeps the copies in sync	§7.12
Repository	Spring Data interface implemented at startup (`JpaRepository`, `CrudRepository`)	§8.8
`REQUIRES_NEW`	propagation that suspends the caller's transaction and runs its own	§7.9
Resilience4j	library for circuit breaker, retry, rate limiter, bulkhead, time limiter	§16.8
Resource server	API that accepts access tokens (`oauth2ResourceServer`)	§11.3
`ResponseEntity`	full control of status, headers and body in MVC	§5.4
`ResponseEntityExceptionHandler`	advice base class mapping Spring MVC exceptions to `ProblemDetail`	§5.7
REST	style of HTTP API built around resources (URLs) and standard methods and status codes	§5.1
`RestClient`	Framework 6.1 fluent blocking HTTP client	§6.2
`RestTemplate`	legacy blocking HTTP client; deprecation announced	§6.9
`RestTestClient`	Framework 7 fluent test client for MockMvc or a real server	§12.9
Retry	calling an operation again after a transient failure, usually with back-off and jitter	§4.7
`Retryable`	Framework 7 retry annotation (1 + `maxRetries` attempts)	§4.7
`RetryableTopic`	spring-kafka non-blocking retries via retry topics	§14.15
`RetryTemplate`	programmatic retry (Framework 7 `org.springframework.core.retry`)	§4.7
Role	authority whose name starts with `ROLE_`	§10.6
Role hierarchy	`ROLE_ADMIN > ROLE_USER`: higher roles include lower ones	§10.6
Rollback-only	flag on a shared transaction that forces a rollback at commit	§7.9
Rollback rules	unchecked → rollback, checked → commit by default	§7.10

S

Term	Meaning	See
Saga	sequence of local transactions across services, with compensating steps that undo earlier ones on failure	§14.17
Salt	random value mixed into each password before hashing so equal passwords give different hashes	§10.5
SAML	older XML-based single sign-on standard used by enterprises; Spring Security supports it	§11.18
Sampling (tracing)	share of traces recorded; Boot default `0.1`	§15.7
Savepoint	point inside a transaction that can be rolled back to (`NESTED`)	§7.9
`Scheduled`	method run on a timer (`fixedRate`, `fixedDelay`, `cron`)	§14.7
Scheduler (Reactor)	thread pool abstraction: `parallel`, `boundedElastic`, `single`	§13.8
Scope	how many bean instances exist and how long they live	§2.6
Scope (OAuth2)	permission requested for an access token (`orders.read`); mapped to `SCOPE_x` authorities	§11.1
Scoped proxy	stand-in that finds the request/session instance on every call	§2.6
Security headers (HSTS, CSP...)	response headers that make browsers safer: force HTTPS, restrict scripts, block framing	§10.11
`securityMatcher`	which requests a whole `SecurityFilterChain` handles	§10.3
`SecurityContextHolder`	`ThreadLocal` holder of the current `Authentication`	§10.2
`SecurityFilterChain`	ordered list of security filters for matching requests	§10.3
Self-invocation	`this.method()` inside a bean bypasses its proxy and all proxy features	§4.5
Service discovery	how a service finds the current addresses of another (Eureka, Consul, Kubernetes DNS)	§16.4
Service mesh	infrastructure layer of sidecar proxies handling traffic, mTLS and retries between services	§16.1
`ServiceConnection`	Boot takes connection details from a Testcontainers container	§12.14
Servlet / servlet container	Java standard API for HTTP request handling / the server that runs servlets (Tomcat, Jetty)	§5.2
Session (HTTP)	server-side state for one browser, found through the `JSESSIONID` cookie	§10.10
Session fixation	attack reusing a pre-login session id; Spring changes the id at login	§10.10
`SessionCreationPolicy`	`IF_REQUIRED`, `ALWAYS`, `NEVER`, `STATELESS`	§10.10
ShedLock	library that lets a `@Scheduled` job run on one instance only	§14.7
SIGTERM / SIGKILL	operating-system signals: "please stop" (allows graceful shutdown) / "stop now" (cannot be handled)	§15.9
Singleton scope	one bean instance per container, shared by all threads (default)	§2.6
Sinks	Reactor API to push values into a `Flux` / `Mono` from imperative code	§13.9
Slice (test)	partial context for one layer (`@WebMvcTest`, `@DataJpaTest`...)	§12.7
`SmartLifecycle`	beans started/stopped with the context, ordered by phase	§2.7
SMTP	Simple Mail Transfer Protocol: how applications send e-mail	§14.4
SPA	single-page application: a browser app (React, Angular...) that calls APIs from JavaScript	§10.8
Span	one timed operation in a trace, with trace id, span id and parent	§15.7
SpEL	Spring Expression Language: `#{...}` in `@Value`, security and cache expressions	§3.5
Spock	Groovy-based test framework; supported again in Boot 4.1	§17.13
Spring AI	Spring project for LLM chat, embeddings, RAG, tools and MCP	§17.1
Spring Authorization Server	Spring's own OAuth2/OIDC authorization server, now part of Spring Security	§11.14
Spring Batch	framework for large, restartable batch jobs (read-process-write in chunks)	§1.3
Spring Boot	Spring + starters + managed versions + auto-configuration + embedded server + Actuator	§1.2
Spring Cloud	projects for distributed systems: config, discovery, gateway, load balancing	§16.1
Spring Cloud Stream	binder-based messaging abstraction over Kafka, RabbitMQ and others	§14.18
Spring Data	family of repository modules sharing one programming model	§9.1
Spring Data JDBC	aggregate-oriented mapping over plain JDBC, no persistence context	§9.2
Spring Data REST	exports repositories as HAL REST endpoints	§9.3
Spring Framework	the core project: container, AOP, MVC, WebFlux, transactions; Boot builds on it	§1.1
Spring Initializr	start.spring.io project generator	§1.5

Term	Meaning	See
Spring Modulith	project for structuring and verifying a modular monolith	§1.3
Spring Security	Spring's authentication and authorization framework, built from servlet filters	§10.1
Spring Session	stores HTTP sessions in Redis or JDBC so all instances share them	§10.10
SQL	Structured Query Language for relational databases	§7.1
SSE	Server-Sent Events: one-way server push over `text/event-stream`	§13.12
SSL	the old name of TLS; still used in property names (`server.ssl.*`)	§3.14
SSO	single sign-on: log in once and reach several applications	§11.2
Stack trace	list of method calls printed with an exception; read the last "Caused by" first	§5.7
Starter	code-free dependency bringing one capability's libraries and auto-configuration	§1.7
Stateless	keeps no per-client data between requests (no HTTP session); every request carries what it needs	§11.1
`StepVerifier`	reactor-test tool that asserts the signals of a publisher	§13.5
Stereotype annotation	`@Component` and its specialisations `@Service`, `@Repository`, `@Controller`	§2.3
Structured logging	JSON log lines (`ecs`, `logstash`, `gelf`), Boot 3.4+	§15.8
Structured output	converting an LLM answer into a Java type (`.entity(...)`)	§17.7
`SpringBootApplication`	`@SpringBootConfiguration` + `@EnableAutoConfiguration` + `@ComponentScan`	§1.8
`SpringBootTest`	test that starts the whole application context	§12.6
Swagger UI	web page that renders an OpenAPI description and lets you try the endpoints	§5.12
Synchronous	the caller waits until the operation finishes and has its result	§14.2

T

Term	Meaning	See
`TaskDecorator`	wraps submitted tasks to copy `ThreadLocal`s (MDC) to worker threads	§14.4
TCP	Transmission Control Protocol: reliable, ordered byte stream under HTTP, JDBC and Kafka connections	§6.4
Test pyramid	many fast unit tests, fewer slice tests, few full-context tests	§12.2
`TestConfiguration`	configuration for tests that adds to (not replaces) the app's configuration	§12.6
TestContext Framework	spring-test engine: `SpringExtension`, listeners, context cache	§12.5
Testcontainers	library starting real services in Docker for tests	§12.14
Thread / thread pool	a thread runs code concurrently; a pool keeps a fixed set of threads that take tasks from a queue	§14.4
Thread-per-request	classic servlet model: one thread per request from start to end	§13.1
`ThreadLocal`	variable with a separate value per thread; holds the security context and the transaction	§7.7
Throughput	how much work is done per unit of time (requests or messages per second)	§13.15
Time limiter	Resilience4j pattern that cancels a call taking longer than a set time	§16.8
Timeout (connect / read)	time to establish a connection / to wait for response data	§6.4
TL;DR	"too long; didn't read": the short summary at the end of a section	§1.1
Token exchange	OAuth2 grant (RFC 8693) that swaps one token for another, e.g. for a downstream service	§11.10
Token relay	forwarding the user's access token to downstream services	§11.10
Tool calling	the model requests a function; the application runs it (`@Tool`)	§17.8
Topic (Kafka)	named stream of records in Kafka, split into partitions	§14.8
TOTP	time-based one-time password: the 6-digit code from an authenticator app (a second factor)	§10.4
Trace / `traceparent`	a request's tree of spans / the W3C header that propagates it	§15.7
Transaction	group of database operations that succeed or fail together (ACID)	§7.7
`Transactional`	declarative transaction around a method (proxy-based)	§7.7
`TransactionalEventListener`	listener bound to a transaction phase (`AFTER_COMMIT` default)	§7.15
`TransactionTemplate`	programmatic transaction around a lambda	§7.14
Transient entity	new object unknown to the persistence context	§8.4
TTL	time to live of a cache entry; a provider setting, not an annotation attribute	§9.8

U – W

Term	Meaning	See
Unchecked exception	`RuntimeException` or `Error`; need not be declared; rolls a transaction back by default	§7.10
Undertow	embedded web server removed in Boot 4 (no Servlet 6.1 support)	§17.13
`UnexpectedRollbackException`	commit of a transaction already marked rollback-only	§7.9
Unit of work	set of changes tracked and written together; the persistence context is one	§8.4
Unit test	fast test of one class in isolation, usually without Spring	§12.4
URI / URL	address of a resource; a URL is a URI that also says how to reach it (`https://...`)	§5.1
`UserDetailsService`	loads a user by name for password authentication	§10.4
UUID	128-bit random-looking unique id, often used as a primary key or message id	§8.3
Validation groups	subsets of constraints selected with `@Validated(Group.class)`	§5.6
`Validated`	Spring annotation: validation groups; on a class, method validation via proxy	§5.6
Vault	HashiCorp secret store; Spring Cloud Vault loads secrets from it	§3.8
`Version`	optimistic-locking column (JPA and Spring Data)	§8.14
Vector store	database of embeddings for similarity search (`SimpleVectorStore`, PgVector)	§17.9
Virtual threads	cheap JVM-managed threads (Java 21); `spring.threads.virtual.enabled=true`	§13.15, §15.10
Virtual time	`StepVerifier` clock that advances only when the test says so	§13.5
W3C Trace Context	web standard defining the `traceparent` header used to propagate traces	§15.7
Weaving	linking aspects with targets; Spring does it at runtime with proxies	§4.1
`WebClient`	reactive, non-blocking HTTP client	§13.13
WebFlux	Spring's reactive web framework on Reactor Netty	§13.10
`WebMvcTest`	web-layer test slice	§12.7
WebSocket	long-lived, two-way connection between browser and server	§13.12
`WebTestClient`	fluent test client for WebFlux (and MockMvc)	§13.12
WireMock	stub HTTP server for testing clients against canned responses	§6.8

X – Z

Term	Meaning	See
X25519 / ECDHE	elliptic-curve key-exchange methods TLS uses to agree on a fresh session key	§11.18
XSRF-TOKEN	cookie name in which Spring sends the CSRF token for JavaScript clients to echo back in a header	§10.8
XSS	cross-site scripting: attacker JavaScript running in your page, able to read anything scripts can	§10.11
YAML	indentation-based configuration format (`application.yaml`), an alternative to `.properties`	§3.2

Chapter TL;DR

- **Versions**: Boot 4.1.1 → Framework 7.0.9, Security 7.1.1, Hibernate 7.4, Jackson 3.1, JUnit 6, Java 17+ (25 used); Spring Cloud 2025.1, Spring AI 2.0 and springdoc 3 are versioned separately (§19.2).
- **Annotations** only work with their switch: `@EnableAsync`, `@EnableScheduling`, `@EnableCaching`, `@EnableMethodSecurity`, `@EnableResilientMethods`, `management.observations.annotations.enabled` — and only through the proxy (§19.3).
- **Starters** are modular in Boot 4: `-webmvc` (not `-web`), `-aspectj` (not `-aop`), `-security-oauth2-*`, `-restclient`, `-webclient`, `-flyway`, plus a `-test` starter per technology (§19.4).
- **Defaults worth remembering** (✔): Hikari pool 10 / 30 s wait, `@Async` pool 8 threads `task-`, one scheduler thread, 30 s shutdown phase, only `health` exposed, tracing sampling 0.1, OSIV on, graceful shutdown on (§19.5).
- **Transactions** (✔): `REQUIRED` joins, `REQUIRES_NEW` survives the caller's rollback, `NESTED` = savepoint, `MANDATORY` / `NEVER` throw; checked exceptions **commit**; a caught inner failure → `UnexpectedRollbackException` (§19.6).
- **Beans** (✔): singleton shared, prototype new per lookup and never destroyed; `@Qualifier` > `@Primary` > `@Fallback` > name; init `@PostConstruct` → `afterPropertiesSet` → `initMethod` (§19.7).
- **Security 7**: lambda DSL only; `hasRole("X")` = `ROLE_X`; 401 unknown vs 403 forbidden; CSRF on for cookies, `csrf.spa()` for SPAs; `http.cors()` before authentication (§19.8).
- **Tests**: unit → slice → a few `@SpringBootTest`s; `@MockitoBean` replaced `@MockBean`; explicit `@AutoConfigure...` for test clients in Boot 4; context cache max 32 (§19.9).

- **Choosing**: `RestClient`/HTTP interfaces for blocking calls, MVC + virtual threads for CRUD, WebFlux for streaming, JPA vs `JdbcClient` vs Data JDBC by model, Caffeine local vs Redis shared, Kafka log vs RabbitMQ queues, Resilience4j for circuit breakers (§19.12).
- **Upgrading**: 3.5 first, OpenRewrite, rename starters and properties, Jackson 3 imports, test starters, `@MockitoBean`, lambda-only security (§19.13).

This is the last chapter. Where to go from here:

- Spring Handbook front page — the chapter list, reading paths and how to run the code.
- Back to the start: Chapter 1 — The Spring Ecosystem and Your First App; revision questions: Chapter 18.
- Java Handbook — the language underneath Spring (records, generics, lambdas, concurrency and virtual threads).
- SQL Handbook — the queries, indexes and transaction isolation behind Chapters 7–9.
- Microservices patterns notes — gateway, discovery, circuit breaker, saga, CQRS and outbox, the designs Chapter 16 implements.